SARNas: A Hardware-Aware SAR Target Detection Algorithm via Multiobjective Neural Architecture Search

Wentian Du, Jie Chen, *Member, IEEE*, Chaochen Zhang, Po Zhao, Huiyao Wan, Zheng Zhou, Yice Cao, Zhixiang Huang, *Senior Member, IEEE*, Yingsong Li, *Senior Member, IEEE*, and Bocai Wu

***Abstract*— Most of the existing deep learning-based synthetic aperture radar (SAR) target detection algorithms rely on manual experience to repeatedly adjust structures and parameters to design models suitable for specific scenarios or tasks. The implementation of the above methods is complicated, the design efficiency is low, and it is difficult to ensure the balance between accuracy and complexity. We innovatively propose a hardware-aware SAR target detection algorithm via multiobjective neural architecture search (NAS), referred to as SARNas. First, we design a flexible and efficient search space, a supernet search strategy, and a subnet contribution evaluation strategy. Furthermore, we construct a new NAS loss function, called SARMI-Loss, to guide the learning of an SAR object detector that balances accuracy and computational complexity. Our SARNas method can address the resource limitations of edge devices and automatically search for the optimal SAR target detector in an end-to-end manner for any deep learning-based SAR baseline model. A series of comparative experiments on three SAR image object detection datasets (SSDD, HRSID, and MSAR) demonstrate the superiority of our method. The experimental results with YOLOV5 as the benchmark model show that the detection accuracy of the target detection networks automatically found by using the SARNas method on the SSDD, HRSID, and MSAR datasets can reach 98.5%, 92.8%, and 91.8% in mean average precision (mAP) with only 2.31M, 1.99M, and 2.21M parameters, respectively. The number of model parameters is reduced by 88.9%, 90.46%, and 68.5%, respectively, and the inference speed is increased by 51.6%, 46.1%, and 13.9% without losing accuracy.**

***Index Terms*— Deep learning, neural architecture search (NAS), SAR target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) [2] is an active Earth observation system that achieves Earth observation by transmitting microwaves and collecting reflection information. Due to its unique imaging mechanism, SAR has stronger observation capabilities than optical remote sensing and can realize all-day and all-weather observations. At the same time, SAR has a certain penetration ability, and strong antijamming and camouflage recognition abilities. Because of the above advantages, SAR systems are widely used in both civilian and military fields.

In the manual processing of SAR images, the massive scale of the available data information and image features in comparison to optical imaging greatly affects the processing speed and accuracy. At the same time, many SAR image processing tasks, especially military tasks, demand a high level of speed and accuracy for SAR image recognition. In this context, researchers began to study SAR target detection algorithms, hoping to replace manual work with automated detection of the targets contained in SAR images [1]. In early research on SAR image target detection methods, traditional image processing algorithms based on image features were mainly used. These methods include target detection algorithms based on structure features [3], gray features [4], and texture features [9], [10]. The constant false alarm rate (CFAR) method based on gray features was popular in early SAR target detection, and a series of improved methods [7], [11], [12] based on CFAR achieved notable results. These algorithms have the advantages of simple design and requiring only a small amount of computation, but they focus only on the statistical distribution characteristics of SAR images while ignoring the more abundant semantic feature information of SAR targets, and they have low detection accuracy and poor generalization ability against complex backgrounds with environmental clutter and strong interference. In addition, some feature detection methods based on manual design have been studied [13], [14], [15]. Those methods rely on the artificial design and extraction of SAR image features and perform well only for specific types of scenes. Similarly, they fail to realize the design requirements of high efficiency and strong generalization for SAR target detection algorithms.

In recent years, with the continuous improvement of the computing power of hardware devices, convolutional neural network (CNN) models based on deep learning have gradually begun to emerge for the detection of objects in optical images. A CNN is a deep learning model that can effectively extract features of 2-D image data. Its advantage is that it can replace artificial feature design and automatically carry out active learning and extraction of image features. At the same time, the multiple convolutional layers of a CNN enable

Manuscript received 28 January 2023; revised 29 May 2023; accepted 3 July 2023. Date of publication 5 July 2023; date of current version 8 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62001003, in part by the Natural Science Foundation of Anhui Province under Grant 2008085QF284, and in part by the China Postdoctoral Science Foundation under Grant 2020M671851. *(Corresponding authors: Jie Chen; Zhixiang Huang.)*

Wentian Du, Chaochen Zhang, Po Zhao, Huiyao Wan, Zheng Zhou, Yice Cao, Zhixiang Huang, and Yingsong Li are with the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, Hefei 230601, China (e-mail: p21301163@stu.ahu.edu.cn; p22301204@stu.ahu.edu.cn; Zp04169013@163.com; p22101006@stu.ahu.edu.cn; zhouzheng19961123@163.com; yccao@ahu.edu.cn; zxhuang@ahu.edu.cn; liyingsong@ieee.org).

Jie Chen is with the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, Hefei 230601, China, and also with the 38th Research Institute, China Electronics Technology Group Corporation, Hefei 230088, China (e-mail: jiechen@ustc.edu).

Bocai Wu is with the 38th Research Institute, China Electronics Technology Group Corporation, Hefei 230088, China (e-mail: 18110995593@189.cn).

Digital Object Identifier 10.1109/TGRS.2023.3292618

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

such a model to perform more complex image recognition functions than traditional image processing algorithms. Thus, it can better fit the image target features to be learned. CNN models have achieved very good results for target detection in optical images. Classical CNN models for target detection include the two-stage models: R-CNN [16], Fast R-CNN [17], and Faster R-CNN [18], as well as the single-stage YOLO series [19], [20], [21], [22] and SSD [23].

Because of the excellent performance of CNNs in optical image detection tasks, the CNN method has been gradually introduced into research on SAR target detection to solve the problems of the low detection rate and poor robustness of traditional SAR image processing methods. The main idea of these efforts is to modify a CNN model for optical image detection to obtain a new CNN model for SAR image target detection by considering the unique properties of SAR images. For example, on the basis of Faster-RCNN, Wang et al. [24] improved the feature candidate regions for SAR image detection to improve the detection effect. Liang et al. [25] modified the loss function on the basis of SAR image characteristics to improve the detection performance of a YOLO-based CNN model on high-resolution SAR images. Wang et al. [26] improved the SSD model and proposed a data expansion and transfer learning method for SAR images, which improved the training effect of the developed CNN model on small-scale SAR image datasets with complex backgrounds.

Although deep learning models have achieved good results in SAR target detection, from the perspective of model design, the following difficulties remain.

1) The design of a deep neural network relies on manual effort and expert experience. To build a satisfactory deep neural network for SAR target detection, continuous trials and adjustments are inevitable. This process is cumbersome and time-consuming, and the final model is usually only a locally optimal one.
2) Hardware platform deployment with the limited computing resources available for edge applications requires customized model compression strategies for computing, which often makes it difficult to ensure a suitable balance between precision and computational burden.

Most of the currently popular target detection algorithms (including optical and network models dedicated to target detection in SAR images) do not specifically consider lightweight design for edge device deployment. For example, the classical target detection algorithm Faster R-CNN [18] has 41.12M model parameters, 91.41 G floating-point operations (FLOPs), and an average inference time of 45.36 ms for each SAR image. The network model FBR-Net [2] designed for SAR target detection has 32.5M model parameters and an average inference time of 40.1 ms. However, current mainstream edge devices have low memory capacity and computing power. Current mainstream AI edge deployment devices, such as the Nvidia Jetson Nano [a single-board computer (SBC)], have only 2 GB of storage space and a computing power of 472 GFLOPs per second (FLOPS). This means that, once other basic software and environments have been loaded on such an edge device, the remaining computing resources available for an object detection model will be insufficient to support corresponding edge reasoning applications. Obviously, from the perspective of the practical application of SAR target detection algorithms, the requirements and challenges of the lightweight detection models needed in the case of edge deployment cannot be ignored.

To realize the goal of autonomous neural network design, automated machine learning (AutoML) has been widely studied in recent years. AutoML is an automated data-driven method that can make a series of machine learning training decisions using only provided training data [27]. Among such methods, neural architecture search (NAS) [28] is an important subfield of AutoML. NAS efficiently evaluates structural performance and finds the best network structure in a massive set of neural network structures. Since NAS was first proposed in 2017, a series of classic works [29], [30], [31], [32], [33] have shown amazing results in the use of NAS for the automatic design of CNN models. The CNN structure obtained by the NAS method is often very different from the CNN designed manually, but it can still achieve performance as good as manually designed networks. This indicates that the NAS method can extract a model structure with excellent performance without manual intervention and, thus, play an important role in the CNN model design task. Therefore, in recent years, a small number of SAR target detection studies based on NAS have emerged [40], [41], [42]. These methods focus on the idea of indirect search, using SAR classification datasets to search for a network structure and then using the search results as part of an SAR target detection model, such as the backbone. Nevertheless, the current approach of directly embedding the classification backbone found by NAS into the target detector for fine-tuning leads to error accumulation, resulting in failure to effectively solve the problem of local optimality. Moreover, the backbone found in this way may not be applicable to a different specific SAR object detection dataset because of inconsistent evaluation metrics. In contrast, the approach applied in this article is to directly use the SAR target detection dataset in the search and consider the actual deployment of the model on an edge device to optimize the search, which is fundamentally different from previous work.

Therefore, to meet the computational resource constraints of edge applications and improve the automation level of deep model design for SAR target detection, inspired by the idea of NAS, we innovatively propose an SAR target detector construction strategy based on multiobjective NAS, referred to as SARNas. Our SARNas method is guided by a joint objective based on the target detection accuracy and computational complexity of the model to automatically conduct an end-to-end search for the optimal SAR target detector. Furthermore, our SARNas method can be embedded as a plug-in framework into arbitrary deep learning object detectors. The main contributions of this article are summarized as follows.

1) We innovatively design a flexible and customizable efficient search space, a supernet search strategy, and a subnet contribution evaluation strategy. This enables SARNas to autonomously perform an end-to-end search for an optimal SAR object detector that satisfies both accuracy and computational complexity constraints based on any state-of-the-art object detector.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

2) To accommodate the resource constraints of edge applications, we consider target detection accuracy and model computational complexity (including inference speed, parameters, and FLOPs) as joint objectives and design a new NAS loss function, SARMI-Loss, to guide the search for SAR target detectors with more balanced performance. In addition, we verify the effectiveness of our method by taking the resource capacity of typical edge AI devices as the maximum upper bound constraint on the FLOPs.
3) Our SARNas method can be used as an optimization tool to automatically optimize any deep learning SAR target detector in accordance with the computing resource capacity of a given edge device, thus enabling extremely strong flexibility and room for customization. In other words, the maximum upper limit on computational complexity (FLOPs and so on) and efficient search blocks in the search space can be freely set and replaced in accordance with application requirements.

II. Related Work

A. SAR Target Detection Algorithms Based on Deep Learning

Due to the excellent performance of deep learning technology in image classification and target detection tasks, many SAR target detection algorithms based on deep learning technology have emerged in recent years. At present, the most popular target detection algorithms are based on optical image datasets. However, due to their special imaging mechanism, SAR images have many differences in image features compared with optical images. For example, in contrast to optical images, SAR images contain only one band of echo information. Moreover, the SAR imaging effect is closely related to the SAR target medium, water content, and roughness, and these factors will interfere with the SAR image target detection task to varying degrees. Consequently, a target detection algorithm with excellent performance on optical images cannot necessarily directly achieve the same excellent effect in SAR target detection.

Therefore, SAR image target detection algorithms based on deep learning are generally designed based on an existing target detection model in combination with the properties of SAR images in order to propose a CNN model structure suitable for SAR image detection tasks. Based on YOLOX [35], Wan et al. [34] proposed an anchor-free SAR target detection algorithm considering the multiscale, strong scattering, and sparsity characteristics of SAR targets. Xia et al. [36] introduced a vision transformer [37] into a CNN-based SAR target detection network to improve the global information perception ability for the SAR target detection task. Feng et al. [38] designed a position-enhanced attention module that can be applied to a CNN model for SAR images to guide the detection algorithm to better focus on the areas where SAR targets are located. It effectively alleviates the problem of recognition accuracy degradation caused by the strong scattering and complex backgrounds of SAR images.

B. Neural Architecture Search

Because the design of neural networks is highly dependent on expert experience and requires many rounds of verification and modification, the design cost of CNN models is high. Thus, AutoML has attracted much attention to reduce the burden on designers and explore neural network architectures that can surpass those based on human design. As an important technical branch of AutoML, NAS has performed well in the automatic design of neural network structures in recent years.

Zoph et al. [28] proposed the NAS method for the first time and designed a component module "cell" of a CNN network based on reinforcement learning. A network model consisting of such stacked "cells" achieved good results on ImageNet. To reduce the computational cost of NAS, Pham et al. [39] proposed a method called ENAS, which improves the efficiency of NAS by sharing parameters among the substructures of NAS during the training process. Liu et al. [29] developed a new approach to the NAS method by making the discrete search space continuous to realize gradient optimization processing for neural network search problems.

With the increasing maturity of NAS methods, the predominant direction of NAS research is gradually approaching practical applications. Related work has begun to consider how to combine NAS with an actual neural network deployment hardware platform for practical applications. ProxylessNAS [33] combines NAS with model compression to enable the use of the NAS method while reducing memory consumption on the GPU. MnasNet [32] and FBnet [31] have been proposed as a type of NAS method with a multiobjective reward mechanism considering the practical problems faced in the deployment and application of neural networks (such as limited computing resources of some devices and slow inference speed) and additionally consider the accuracy and inference delay of neural network models in the structure search process.

C. SAR Target Detection Model Design With NAS

At present, some works have applied NAS to search for a suitable network structure for SAR image processing tasks. Zhu et al. [40] used the NAS method to assist in the design of a neural network for SAR image classification. Li et al. [41] used differentiable structure search technology based on the DARTS method to search for feature aggregation structures to be used in SAR ship target detection tasks. Zhang et al. [42] proposed FL-DARTS (a modified version of DARTS) to search for a network structure based on an SAR image classification dataset, used the structure obtained in the search as the backbone feature extraction network of an SAR target detection algorithm, and finally achieved good results in the SAR target detection task.

III. Methods

This section introduces the specific underlying concepts of the proposed SARNas method, which builds a multiobjective NAS strategy to automatically search for SAR target detectors in an end-to-end manner.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

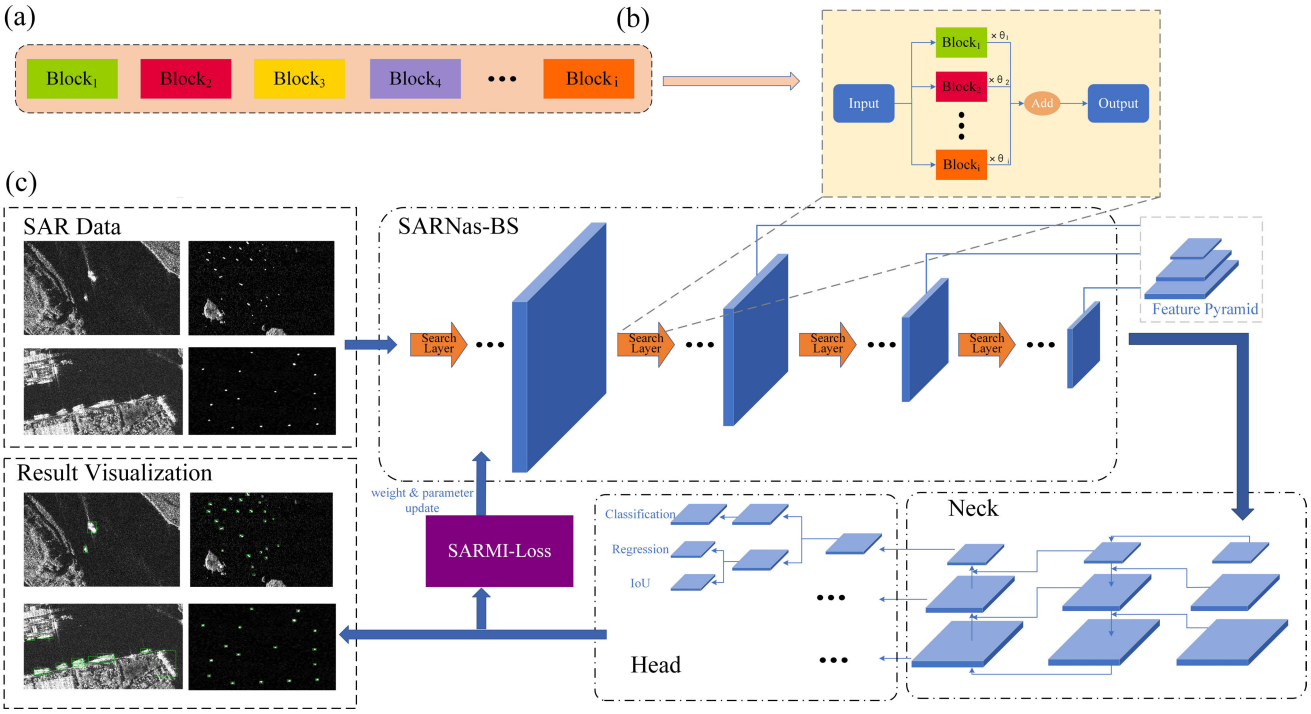


Fig. 1. Overall structure of the SARNas method. (a) Search space. (b) SearchLayer. (c) SARNas pipeline.

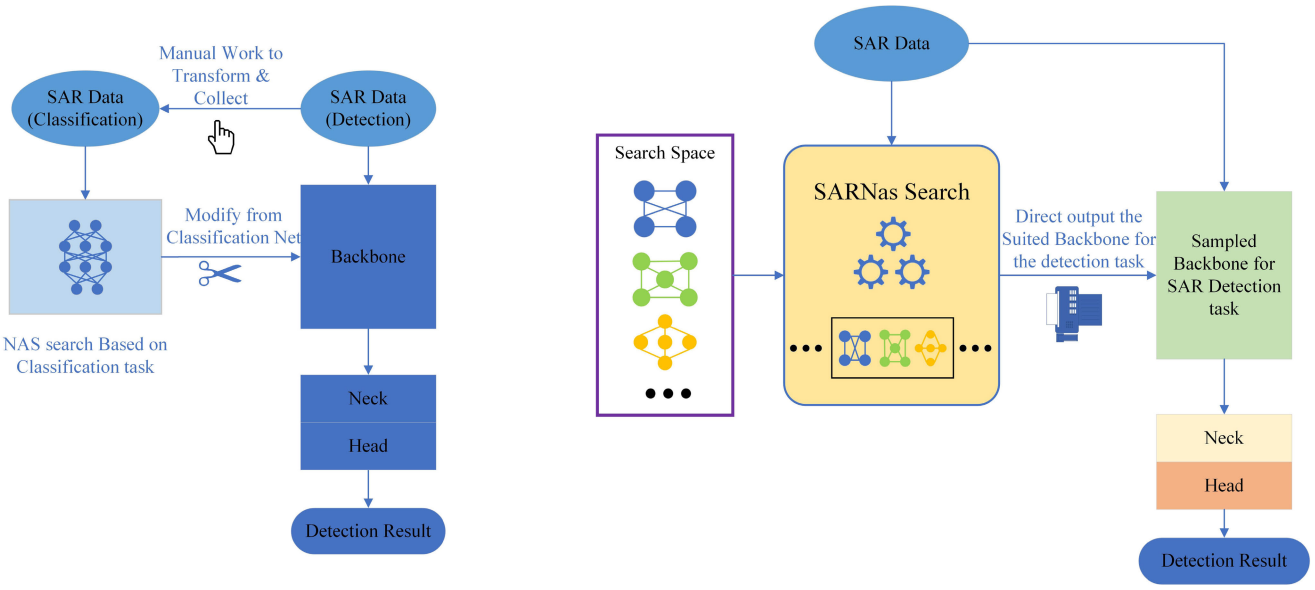


Fig. 2. Process comparison of (left) traditional NAS application mode and (right) our proposed NAS method SARNas.

A. Overall Process of SARNas

The overall structure of the SARNas method is shown in Fig. 1. The overall process of SARNas is similar to that of common object detection frameworks and is divided into three main parts: SARNas-backbone searcher (SARNAS-BS), the Neck and Head parts, and SARMI-Loss (an SAR multi-indicator loss). After SAR image data are input into SARNas, SARNAS-BS first performs feature extraction; then, the extracted features are input into the subsequent Neck and Head parts to transform the model training results; and finally, SARMI-Loss is used as the loss function to calculate the loss value. The loss value is then passed back to SARNas-BS to guide the network structure search.

As shown in the left part of Fig. 2, some previous NAS methods for object detection separate the backbone network search from the object detection model training process; that is, they use classification datasets to train a classification network separately, and the classification network found in this way is then manually incorporated into the object detection network to be improved.

As shown in the right part of Fig. 2, the idea of our proposed SARNas method is to build a supernet method SARNas-BS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

responsible for the backbone search task and combine it with the Neck and Head parts of the object detection network to form an overall search network for the backbone structure search. To reduce the amount of NAS computation and search time, we refer to the design ideas of DARTS [29] and FBnet [31], and use a one-shot NAS method to construct SARNas-BS. In the search process, the dataset is directly imported to train the supernet, and after training is complete, the supernet is sampled to obtain the optimal subnetwork architecture, which is used as the search result of our proposed SARNas method.

Notably, the backbone NAS supernetwork SARNas-BS is obtained by stacking multiple SearchLayers (details of the SearchLayer design are given in the following). SARNas is a search network that can be trained directly using object detection datasets, similar to an ordinary object detection network. Therefore, in comparison to the traditional method depicted in the left part of Fig. 2, the additional work of finding or making SAR image-related classification datasets can be avoided, and the SAR target detection datasets can be directly used for training. At the same time, the final sampling output of the SARNas search network is also suitable to become the backbone structure to be combined with the Neck and Head modules currently adopted for target detection. The backbone network can be directly combined with these subsequent modules to form a target detection network to be trained on the target detection task in SAR images without additional structural modifications.

When performing the NAS task, SARNas can directly guide the model weight update based on the loss function, similar to the typical object detection network training process. The difference between the two is that the parameters of the SARNas search network include the parameters of the sought model structure; that is, as the loss value decreases during the training of the NAS network, the search results of SARNas become more suitable for the current SAR target detection task in combination with the Neck and Head modules.

B. Plug-in NAS Method: SARNas-BS Search Space

As shown in Fig. 1(a), the search space of SARNas includes several encapsulated computational modules named blocks. As shown in Fig. 1(b), these computational modules participate in parallel computations in each SearchLayer, and eventually, each SearchLayer determines a certain computational module in the search space as its search result. Therefore, as a basic part of the search process, the types of blocks involved in the search will directly affect the final model sampling results. In this article, to optimize the search space, more efficient computational block types are introduced. We refer to [43], [44], and [45] to design the computational block structures of SARNas, as shown in Fig. 3. We choose these three types of blocks as our baseline structures for the following reasons.

The core concern of this article is to design a suitable NAS method to automatically search for a balanced deep learning architecture compatible with the limited computing resources of embedded application platforms. The three types of blocks considered in this article are the most cutting-edge lightweight blocks in the field of deep learning and are well representative of the capabilities of deep learning models. On this basis, we hope to search out a highly applicable lightweight backbone for SAR target detection tasks through the NAS method, and these three blocks are simply the benchmark architectures selected to verify our proposed method. More generally, our method can be used as a plug-in framework to automatically search for any kind of block.

1) Inverted Residual Block: Fig. 3(a) shows the structure of the inverted residual block. It contains three convolutional layers, including pointwise (kernel size: 1×1) convolution and depthwise (kernel size: $k \times k$) convolution, and a residual structure is adopted in the middle. The parameter e stands for "expand" and represents the multiple by which the number of feature channels is expanded inside the inverted residual block.

The inverted residual block was proposed in MobileNetV2 [43] and is an innovative improvement achieved by introducing inverted residuals on the basis of a depthwise separable convolution module. In the process of SAR image detection, the inverted residual structure can effectively alleviate the phenomenon of information loss caused by changes in the dimensions of the SAR image feature map during the convolution operation. In recent years, the designs of several SAR target detection networks based on deep learning (such as [34], [60], and [61]) have referenced the design of inverted residual modules, and good results have been achieved.

2) Ghost Module Block: Fig. 3(b) shows the structure of the Ghost module block, which includes an ordinary convolutional layer (convolution kernel size: $k \times k$) and depthwise convolution, where the parameter r stands for "ratio" and represents the reduction factor. The number of output channels of the module, H_{out}/r, is equal to the number of channels of the residual structure inside the Ghost module block.

The Ghost module, which was proposed in [45], performs a convolution operation with the advantages of lightweight and high efficiency. When a CNN performs feature extraction operations on an input SAR image, it often produces similar feature maps, that is, redundant information, leading to invalid and redundant calculations and thus unnecessarily increasing the workload of model execution. The Ghost module concatenates the intrinsic feature maps and the Ghost feature maps obtained via the convolution operation at the channel level, thereby avoiding the most redundant calculations. At the same time, it ensures the information diversity of the SAR image feature map.

3) RepVGG Block: Fig. 3(c) shows the structure of the RepVGG block, which performs 3×3 and 1×1 convolution calculations on the input feature maps and then sums the two sets of results. The SE module [57] in Fig. 3(c) represents an additional attention mechanism introduced after the block calculation, and the use of the SE module is optional. In the subsequent search space, many computational block variants of these three types of blocks can be obtained by adjusting K (convolution kernel size), e (expansion factor), r (reduction factor), and whether the SE module is used.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OUR PROPOSED SARNAS ALGORITHM

Algorithm 1: The process of our proposed SARNas method

Create a SARNas-BS supernet $\mathcal{A}$ as a combined series of $SearchLayer$s
w: the weights of $\mathcal{A}$
θ: the coefficient set of all blocks in these $SearchLayer$s
$\mathcal{L}_{precision}$: the loss function that contains only the precision loss value
$\mathcal{L}_{SARMI}$: the loss function that contains multiple loss terms

// stage 1. Preliminarily train the weights of the NAS supernet
while (stage 1 not end) **do**
  Update the weights w by descending $\nabla_w \mathcal{L}_{precision}(pred, label)$
end while
// stage 2. Alternately train the weights w and block coefficients
while (not converged) **do**
  1. Update the weights w by descending $\nabla_w \mathcal{L}_{precision}(pred, label)$
  2. Update the coefficients θ by descending $\nabla_\theta \mathcal{L}_{precision}(pred, label)$
end while
  // Use the loss function $\mathcal{L}_{multi-indicator}$ to guide the search direction
for (last epochs) **do**
  1. Update the weights w by descending $\nabla_w \mathcal{L}_{SARMI}(pred, label)$
  2. Update the coefficients θ by descending $\nabla_\theta \mathcal{L}_{SARMI}(pred, label)$
end for

// Get the search result
Step 1: Sample the best blocks of each $SearchLayer$ according to the results for θ
Step 2: Combine every block sampled in step 1 to obtain the resulting architecture a
Step 3: Substitute a for the backbone of the original object detection CNN

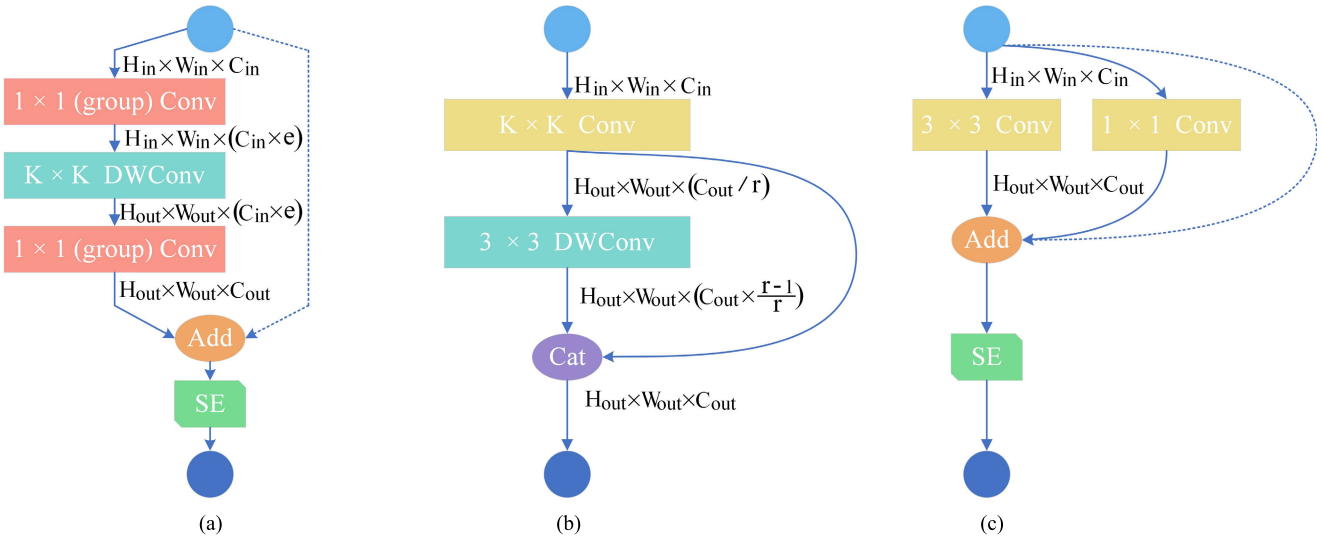


Fig. 3. Structure diagrams of three kinds of computational blocks used in our proposed SARNas method. (a) Inverted residual block. (b) Ghost module block. (c) RepVGG block.

RepVGG was proposed in [44] and has an innovative multibranch convolutional network structure. Some popular network components with excellent performance have been proposed in recent years, such as depthwise separable convolutions and grouped convolutions. These methods can be used to effectively realize a lightweight model network, but its memory access cost (MAC) will also increase significantly, which will lead to a slowdown of the model. The RepVGG block adopts a convolution calculation structure with a multibranch topology, which can effectively decouple the feature information of SAR images and effectively improve the efficiency and accuracy of model training and inference.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

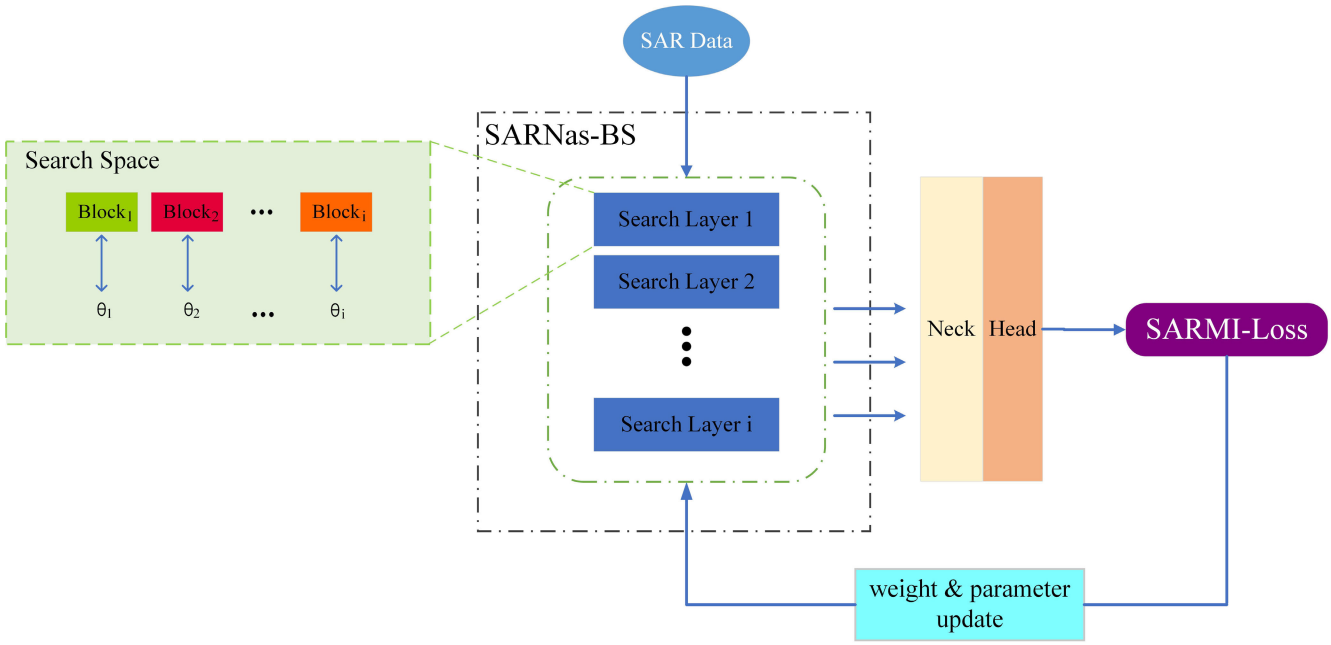


Fig. 4. Pattern of stacking the SearchLayers in SARNas-BS.

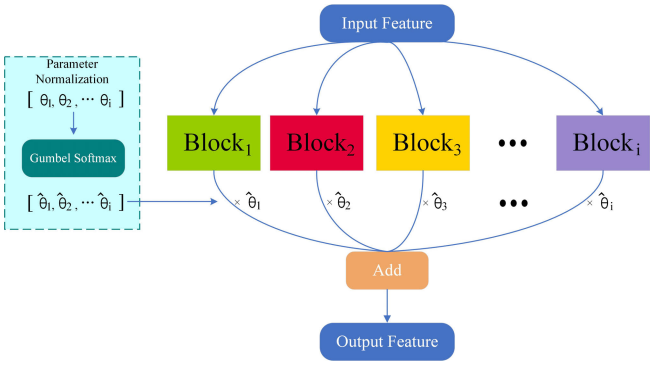


Fig. 5. Internal structure of a SearchLayer.

C. Search Strategy

SARNas-BS consists of a certain number of SearchLayers, in which each layer adds multiple candidate computational blocks in parallel, as shown in Fig. 4. In each SearchLayer, we assign coefficients θ to each candidate computational block. Different coefficients θ are assigned to distinguish the importance of different computational blocks in each SearchLayer. At the same time, this coefficient information will also participate in the process of supernet training and affect the execution effect of the supernet through the different coefficient values. We design these coefficients to be updated by backpropagation during training.

As shown in Fig. 5, in the calculation process of each SearchLayer, when the feature map from the previous layer is passed into the SearchLayer, the feature map is passed to all candidate computational blocks in the SearchLayer for parallel calculation. Moreover, the coefficient θ_i corresponding to each candidate's computational block Block_i is normalized using the Gumbel softmax distribution [58] to obtain the actually calculated coefficients $\hat{\theta}$, whose sum of values is fixed to 1. The Gumbel softmax calculation formula is given as follows:

$$\hat{\theta}_i = \frac{\exp\left(\frac{\theta_i + g_i}{\tau}\right)}{\sum_{j=1}^{k} \exp\left(\frac{\theta_j + g_j}{\tau}\right)}, \quad \text{for } i = 1, \ldots, n \tag{1}$$

where $g_i \sim \text{Gumbel}(0, 1)$ is random Gumbel noise and τ, which is also known as the temperature parameter, regulates the density of y_i after reparameterization.

By introducing the Gumbel softmax calculation, we can backpropagate the block coefficients θ and update them. Finally, we multiply the calculation results of each candidate computational block $\text{Block}_{1\sim i}$ by its corresponding actual operation coefficient $\hat{\theta}_{1\sim i}$ and sum the outcomes. Thus, we obtain the final output feature map of the SearchLayer. The formula is given as follows:

$$\text{LayerOutput} = \sum_i \hat{\theta}_i * \text{Block}_i$$

where

$$\hat{\theta}_i = \text{Gumbel Softmax}(\theta_i), \quad i = 1, \ldots, n. \tag{2}$$

We distinguish the computational importance of each candidate computational block for the feature map passed to it in the SearchLayer by the values of the coefficients θ_i. As the training of the supernet continues, the computation coefficients θ of each SearchLayer are also updated according to the loss function. Finally, after training, the coefficients θ corresponding to the candidate computational blocks contained in each SearchLayer will be extracted. At this time, we consider $\text{Block}_{\arg\max_i \theta_i}$, whose coefficient $\theta_{\max}$ has the maximum value among all θ values of its SearchLayer, to be the optimal block $\text{Block}_{\text{best}}$ for that SearchLayer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

D. Target Search Guidance Mechanism for Multiobjective Optimization: SARMI-Loss

For the actual deployment of an SAR target detection model, in addition to achieving good detection accuracy, it is also necessary to optimize the comprehensive properties of the model by considering its performance on the actual hardware equipment. Therefore, in the design of our method, in addition to the accuracy, we introduce several major CNN model properties as important indices for the evaluation of the searched network models, such as inference speed, parameter quantity, and computational burden (FLOPs), as reference indicators for the search direction of the SARNas method.

In accordance with the training characteristics of deep learning models, we choose to redesign the loss function by incorporating the network model accuracy, inference speed, parameter quantity, and calculation amount into the calculation of the loss function. Thus, a new loss function called the SAR multi-index loss (SARMI-Loss) is proposed to guide the search direction of SARNas by backpropagating the loss value during supernet training. The formula of SARMI-Loss is given as follows:

$$L_{\text{SARMI}} = \text{precision}_{\text{loss}} + \text{latency}_{\text{loss}} + \text{param}_{\text{loss}} + \text{flops}_{\text{loss}} \tag{3}$$

where $\text{precision}_{\text{loss}}$ represents the current accuracy loss value of the model and is also the original loss function of the object detection model to be improved by SARNas. $\text{latency}_{\text{loss}}$ represents the model inference time loss and reflects the magnitude of the inference time of the current model found in the search. $\text{flops}_{\text{loss}}$ represents the computational burden loss of the current model, which reflects the amount of calculation needed in the model. $\text{param}_{\text{loss}}$ represents the parameter quantity loss of the current model, which reflects the number of parameters of the model.

For each loss value in SARMI-Loss, taking the inference time loss term $\text{latency}_{\text{loss}}$ as an example, we design the calculation pattern as follows. The inference time loss of a SearchLayer is obtained by multiplying the computational latency l_i of every block in the layer by its coefficient $\hat{\theta}_i$ and summing them all. Then, the inference time loss of all SearchLayers is accumulated to obtain the inference time loss $\text{latency}_{\text{loss}}$ of the current model. $\text{flops}_{\text{loss}}$ and $\text{param}_{\text{loss}}$ are calculated following a similar idea. The specific calculation formulas are given as follows:

$$\text{latency}_{\text{loss}} = \beta_{\text{latency}} * \log\left(\sum_{\text{layer}}\sum_{\text{block}} \hat{\theta}_i * l_i\right) \tag{4}$$

$$\text{param}_{\text{loss}} = \beta_{\text{param}} * \log\left(\sum_{\text{layer}}\sum_{\text{block}} \hat{\theta}_i * p_i\right) \tag{5}$$

where the coefficient β is used to coordinate the different loss values during training such that the gap between these loss values will not be so large as to cause an imbalance in the search direction in the NAS process. $\hat{\theta}_i$ represents the weight coefficient of the corresponding candidate computational module Block_i. Similarly, l_i and p_i represent the computational latency and the number of parameters, respectively, of Block_i.

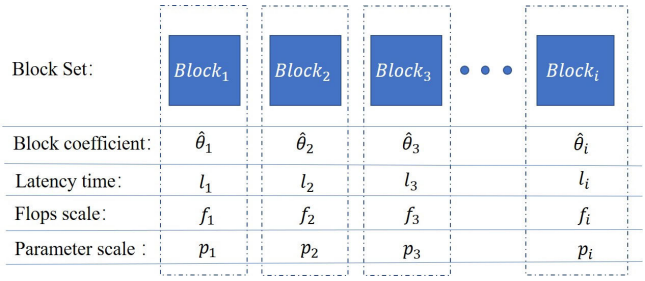


Fig. 6. Schematic of the information list of blocks involved in the calculation of SARMI-Loss.

In actual implementation, an SAR image processing algorithm is often limited by the computing capabilities of the hardware device; thus, the FLOPs of the model should be limited. Inspired by MnasNet [32], we propose the specific implementation idea of adding a new limit coefficient in the computational loss $\text{flops}_{\text{loss}}$, whose formula is given as follows:

$$\text{flops}_{\text{loss}} = \alpha * \beta_{\text{flops}} * \log\left(\sum_{\text{layer}}\sum_{\text{block}} \hat{\theta}_i * f_i\right)$$

where

$$\alpha = \begin{cases} \gamma, & \text{if FLOPs}(m) > T \\ 1, & \text{otherwise} \end{cases} \tag{6}$$

where $\text{FLOPs}(m)$ represents the amount of computation needed in the current network and T represents an upper limit on the amount of computation. α is a parameter whose value corresponds to the FLOPs of the current model, which is used to adjust the value of the computational burden loss accordingly. Specifically, the value of α is γ when $\text{FLOPs}(m) > T$ and is 1 otherwise. In practice, the γ value can be adjusted in accordance with the situation and is usually much larger than 1.

α is a piecewise function that constrains the search direction in accordance with the situation during the SARNas search process. When the number of FLOPs exceeds the expected design limit T, the search focuses on architectures with lower FLOPs. When the network computational burden is within the expected design limit, the search algorithm pays more attention to other performance properties of the model.

As shown in Fig. 6, to enable the calculation of $\text{latency}_{\text{loss}}$, $\text{flops}_{\text{loss}}$, and $\text{param}_{\text{loss}}$, the corresponding latency, FLOPs, and the number of parameters of each candidate block are measured and recorded before the NAS process begins. These values are then retrieved for the calculation of SARMI-Loss during the SARNas search process.

E. Implementation Details of SARNas

During the supernet training process, to ensure that each candidate computational block in each SearchLayer can be fully trained in an objective and fair manner, the detrimental impact of random factors in the search process on the search results should be eliminated. We refer to the training strategies for other NAS methods and propose an NAS training strategy based on the supernet search mode. The training pipeline is described by the pseudocode given in Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

Through experiments, we have formulated an effective NAS strategy, which mainly divides the NAS process into two stages. In the first stage, the overall weights w of the supernet A are trained to convergence; thus, the weights of all candidate computational blocks are involved in the training process to avoid the influence of random factors on the search direction of the NAS process due to premature intervention of the coefficients of the computational blocks. In the second stage, to avoid the possibility of a bilevel optimization problem caused by mutual coupling of the block weights w and coefficients θ involved in the NAS process, we impose alternating updates of w and θ in the supernet using the gradient descent method. During the training process in the first half of the second stage, we use only the loss function $L_{precision}$, which reflects the accuracy, to the first search for a backbone network that can maximize the recognition accuracy. In the second half of the second phase, we replace the loss function guiding the SARNas search with the newly proposed loss function L_{SARMI} for multitask optimization, adding the inference speed, the number of parameters, and FLOPs of the network structure into consideration during the SARNas search. Finally, a sampled architecture result that accounts for both accuracy and ease of hardware deployment is obtained.

IV. Experiments

To verify the effectiveness of our proposed method, we report experimental tests performed on three current mainstream SAR datasets: SSDD, HRSID, and MSAR. In recent years, SAR target detection efforts based on the YOLO series of algorithms [5], [6], [7] have achieved good results. At present, YOLOX [35] and YOLOV5 [46] are the two most representative deep learning object detectors in the YOLO series, with excellent comprehensive performance, and are widely used in engineering applications. Therefore, we take YOLOX as a representative anchor-free detection framework and YOLOV5 as a representative anchor-based detection framework to experimentally verify our proposed method on two popular detection benchmarks in the field of SAR target detection. The experimental results on the above datasets show that our method can effectively balance the inference time, parameter quantity, and FLOPs of the sought model, enabling the feasible performance of the model in actual hardware deployment while ensuring effective recognition accuracy for the SAR target detection task.

A. Datasets

1) SSDD: SSDD is an SAR image ship detection dataset published by Hu et al. [56] in 2017. Since its release, it has become an important benchmark dataset for model performance verification in SAR target detection tasks. The SSDD dataset contains a total of 1160 images and 2456 labeled ship targets, and includes samples captured under different conditions (such as resolution, sample size, sea conditions, and sensor types, all of which have a certain influence on image collection). Because of the various complex conditions under which the sample images were collected, the SSDD dataset provides challenging examples for target identification and detection, and can, therefore, play a useful role in performance evaluation for detection models. More details about the SSDD dataset are given in Table II.

TABLE II
Information of the SSDD Dataset

Dataset Condition	Value(s)
sensors	RadarSat-2, TerraSAR-X, Sentinel-1
polarizations	HH, VV, VH, HV
category	ship
sea conditions	good and bad conditions
resolution	1–15 m
locations	in the sea and offshore
number of images	1160
number of objects	2456

TABLE III
Information of the HRSID Dataset

Dataset Condition	Value(s)
sensors	Sentinel-1 B, TerraSAR-X, TanDEM-X
polarizations	HH, VV, VH, HV
category	ship
resolution	under 3 m
locations	port with cargo and busy canals
number of images	5604
number of objects	16951

2) HRSID: The HRSID dataset is an SAR image ship detection dataset published by Zhang et al. [55] from the University of Electronic Science and Technology of China in 2020. The SAR images in this dataset have high quality and high resolution. At the same time, most of the images in the HRSID dataset contain various factors that can interfere with detection, such as multiple types of ships and complex scenes, which present great challenges for a detection model. The diverse environments and high image quality in HRSID make this dataset convenient for data enhancement and as a benchmark for researchers to evaluate their methods. The details of the HRSID dataset are shown in Table III.

3) MSAR: The MSAR dataset [66] is a large-scale multiclass SAR target detection dataset released by our research team in 2022, which was first published in a paper by Xia et al. [36]. MSAR was constructed based on images collected by the HISEA-1 remote sensing satellite. Compared with general SAR target detection datasets, MSAR is larger in scale and richer in target types. There are 28 449 SAR image slices in the dataset and 60 396 SAR detection targets of four types, namely, aircraft, oil tanks, bridges, and ships. It also involves multiple complex detection backgrounds, such as airports, ports, inshore areas, islands, high seas, and urban areas. This dataset can be used to comprehensively evaluate the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
INFORMATION OF THE MSAR DATASET

Dataset Condition	Value(s)
sensor	HISEA-1
polarizations	HH, VV, HV, VH
categories	ship, bridge, tank, aircraft
locations	land, sea, canal
number of images	28449
number of objects	60396

performance of a model for multiclass SAR target detection tasks. The specific information of the MSAR dataset is shown in Table IV.

B. Experimental Setup and Details

During the experiments, we randomly divided the datasets into two parts, a training set and a test set, which were used for model training and result verification, respectively. Specifically, the ratio of the training set to the test set was 8:2 for SSDD and approximately 6.5:3.5 for HRSID. Our experimental software environment mainly included Python 3.8.12, PyTorch 1.8.0, NVIDIA-SMI driver version 470.74, and CUDA version 11.4. Regarding the experimental hardware and operating system, we used NVIDIA GeForce GTX3070 and Ubuntu 18.04.6 LTS.

In our designed SARNas method, training is divided into two stages, as introduced in Table I. In the first stage, SARNas is used to search for a model structure. In the second stage, the newly obtained model structure is retrained on the dataset to obtain the final training results.

In the first stage, we trained the supernet of SARNas for 90 epochs, and the batch size was set to 2. In the first 15 epochs, only the weights of the supernet were updated, and in the remaining 75 epochs, the weights of the supernet and the block coefficients in all SearchLayers were alternately trained and updated. During this process, only the precision loss function $L_{\text{precision}}$ was used when learning the weights of the supernet; for coefficient learning, only the precision loss function $L_{\text{precision}}$ was used from the 16th epoch to the 55th epoch, and the SARMI-Loss function L_{SARMI} was used from the 56th epoch to the 90th epoch. For the supernet weights, we used the SGD optimizer with the initial learning rate set to 0.01 and the weight decay and momentum set to 0.0005 and 0.9, respectively; for the block coefficients, we used the Adam optimizer [59] with the learning rate set to 0.0001 and the weight decay set to 0.0005.

In the second stage, after the sampled network structure was configured, we set the total training period for the sampled network to 300 epochs, the batch size to 16, and the weight decay and momentum to 0.0005 and 0.9, respectively. Regarding the coefficients in our proposed SARMI-Loss function, we set the default value of α in the $\text{flops}_{\text{loss}}$ term to 1, $\gamma = 10.0$, and β to 0.05, and α was set equal to γ when the FLOPs of the current model found in the search exceeded the upper limit T. The stacked SearchLayer structure of our proposed supernet is shown in Table V.

In Fig. 3, three basic types of blocks are illustrated. The computational blocks used in our experiments were obtained by adjusting these basic types of search blocks. When selecting the search space for our experiment, we set the candidate blocks in accordance with the characteristics of a wide area and various target sizes in the SAR images for detection. For the inverted residual block, we set two large convolution kernel sizes of 5 and 7 and a large channel expansion factor of 6. For the Ghost module and RepVGG blocks, due to their special calculation processes, we referred to the original papers on GhostNet [45] and RepVGG [44] and adopted the convolution size and internal channel factor settings recommended by the original papers in our experiments. The candidate computational blocks considered in the search space are described in Table VI.

C. Evaluation Indices

To verify the specific effect of the backbone network obtained by our search method in combination with the object detection network, we adopt the mean average precision (mAP), a performance metric commonly used in object detection tasks, to measure the model detection performance. According to our experience, the difference in FLOPs does not accurately reflect the difference in inference speed between different models. Therefore, we also adopt the average inference time as a metric to compare the actual efficiency of models in terms of inference time for SAR images.

We use P as an abbreviation for the precision metric and R as an abbreviation for the recall metric, where the precision represents the proportion of correctly detected targets among all detected targets and the recall represents the proportion of correctly identified positive samples among all positive samples. The formulas for P and R are shown as follows:

$$\begin{aligned} \mathrm{P} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{R} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \end{aligned} \tag{7}$$

where TP represents the number of samples that are correctly judged as positive samples by the model, FP represents the number of samples that are incorrectly judged as positive, and FN represents the number of samples that are incorrectly judged as negative.

AP stands for the average precision, which is obtained by calculating the average of the precision results of the detection model at different recall values. The AP index can more comprehensively reflect the target detection ability of a model, and its calculation formula is given as follows:

$$\mathrm{AP} = \int_0^1 \mathrm{P}(\mathrm{R}) dR. \tag{8}$$

AP represents the recognition ability of a model only for targets of a single class, whereas mAP represents the average of the AP values for multiple classes. Its formula is given as follows:

$$\mathrm{mAP} = \frac{\sum \mathrm{AP}}{\mathrm{num}_{\text{classes}}} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE V
SUPERNET STRUCTURE IN SARNAS

Stage	Input	Output	Operator	c	s	n
	$640^2\times3$	$320^2\times32$	Con2d	32	2	1
Stage 1	$320^2\times32$	$160^2\times64$	SearchLayer	64	2	1
Stage 2	$160^2\times64$	$160^2\times64$	SearchLayer	64	1	3
Stage 3	$160^2\times64$	$80^2\times128$	SearchLayer	128	2	1
Stage 4	$80^2\times128$	$80^2\times128$	SearchLayer	128	1	3
Stage 5	$80^2\times128$	$40^2\times256$	SearchLayer	256	2	1
Stage 6	$40^2\times256$	$40^2\times256$	SearchLayer	256	1	3
Stage 7	$40^2\times256$	$20^2\times512$	SearchLayer	512	2	1
Stage 8	$20^2\times512$	$20^2\times512$	SearchLayer	512	1	3

TABLE VI
CANDIDATE COMPUTATIONAL BLOCKS IN THE SEARCH SPACE, WHERE "IR" DENOTES COMPUTATIONAL BLOCKS USING THE INVERTED RESIDUAL STRUCTURE, "GHOST" DENOTES COMPUTATIONAL BLOCKS USING THE GHOST MODULE STRUCTURE, "REPVGG" DENOTES COMPUTATIONAL BLOCKS USING THE REPVGG STRUCTURE, AND THE "SE" COLUMN INDICATES WHETHER THE SE ATTENTION MECHANISM MODULE IS ADDED AFTER THE BLOCK

Block Type	Kernel	Shrink Ratio	Expand Ratio	SE
ir_k5_e6	5	---	6	---
ir_k5_e6_se	5	---	6	✓
ghost_k3_r2	3	2	---	---
ir_k7_e3	7	---	3	---
repvgg_se	---	---	---	✓
skip	---	---	---	---

where num_classes represents the number of classes spanned by all targets.

D. Experimental Results and Analysis

To verify the effectiveness of our proposed method, we designed a series of comparative experiments based on multiple SAR target detection datasets. We selected the currently popular anchor-free detection algorithm YOLOX [35] and the anchor-based algorithm YOLOV5 [46] to conduct SARNas search experiments on two SAR ship target detection datasets, SSDD and HRSID. Finally, we compared the detection effects of the models before and after our search method. In addition, we compared the performance of the new detection models obtained after searching with other popular target detection algorithms and the latest SAR target detection algorithms to prove the effectiveness of our method.

1) Ablation Experiments: To verify that all parts of our proposed NAS method can achieve the corresponding design goals, we conducted a series of ablation experiments on the YOLOX detection algorithm (an anchor-free detection algorithm) to rigorously test the effectiveness of our proposed method.

The ablation experiments that we conducted mainly addressed the following scenarios.

1) We initially considered only the accuracy index (that is, we included only the loss term reflecting the accuracy in the loss function used to guide the search). We used the SARNas search method to search for the most suitable backbone network structure for use with the Neck and Head structure of YOLOX within the search space specified in Table I. The original YOLOX backbone structure was then replaced with the search results sampled after the search.
2) On the basis of the accuracy index, we added $\text{flops}_{\text{loss}}$, introduced in formula (5), to limit the maximum amount of computation of the sought network model. To set this upper limit on the amount of calculation, we investigated commonly used embedded application-specific integrated circuits (ASICs) and chose to take the performance parameters of the Hunxin 3, which is an intelligent computing chip recently released by Anhui Siliepoch Technology Company Ltd., as our experimental reference standard. This chip has a computing power of 350 GFLOPS (350 billion INT8 FLOPS). Considering actual deployment conditions (FP32-bit precision detection model, maintaining a detection rate of 10 frames/s while maintaining 40% computing power for emergencies), we set 5.5 GFLOPs as the upper limit on the amount of calculation required by the sought model.
3) In addition to the accuracy index and limiting the maximum amount of model computation, we further considered the inference time index. That is, following the design of SARMI-Loss as presented in Section III, we added a term calculating the inference time loss $\text{latency}_{\text{loss}}$ to guide the SARNas method to optimize the actual inference time of the sought model.

The performance data from the ablation experiments are shown in Table VII, and the results are visualized in Fig. 7. From the experimental results, we can draw the following conclusions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
RESULTS OF ABLATION EXPERIMENTS

Dataset	Model (ID)	NAS Search Strategy			Params (M)	FLOPs (G)	mAP (%)	Average Inference Time (ms)
		Precision factor	Max FLOPs constraint	Latency factor				
SSDD	YOLOX	—	—	—	5.08	17.57	95.8	11.20
	(1)	✓			2.75	6.72	98.2	8.91
	(2)	✓	✓		1.25	5.26	97.2	7.19
	(3)	✓	✓	✓	1.39	5.45	97.8	5.54
HRSID	YOLOX	—	—	—	5.08	17.71	89.2	10.08
	(1)	✓			1.95	7.09	89.8	8.32
	(2)	✓	✓		1.22	5.10	89.8	7.31
	(3)	✓	✓	✓	1.28	5.46	89.1	4.81

1) Under the search strategy of focusing only on the accuracy index, SARNas combines the candidate computational blocks in the search space and, finally, finds the most suitable network structure for the SAR target detection task. The results can be seen in Table VII, labeled as Model (1). Taking the SSDD dataset as an example, after the original backbone structure of YOLOX is replaced with the backbone structure obtained in the search, our new structure improves the accuracy by 2.4%, reduces the number of parameters from 5.08M to 2.75M, and reduces the FLOPs from 17.57 G to 6.72 G. The inference time is reduced from 11.20 to 8.91 ms. On the HRSID dataset, these indicators are also optimized to a certain extent. Thus, it is proven that the SARNas-BS component of the target detection algorithm proposed in this article can find a backbone structure more suitable for the current SAR image detection task. At the same time, when the search space consists of lightweight calculational blocks, the obtained model can be light in weight.
2) While considering the accuracy index, we additionally restrict the computational burden of the sought structure in accordance with the actual deployment requirements of the detection model. The experimental results for Model (2) in Table VII show that, after the computational burden limit is added, the search for the network structure by SARNas is effectively guided, and the final FLOPs of the search structure are stably controlled within the set upper limit (5.5 GFLOPs). At the same time, the detection accuracy of the model obtained through the search on the HRSID is unchanged, and the detection accuracy on the SSDD dataset is slightly decreased but still higher than that of the original structure of YOLOX. Moreover, with a further decrease in the amount of computation, the inference speed of the model is also accelerated.
3) In the third stage of the ablation experiments, we optimize the actual inference speed of the model by adding the inference time loss to the loss function. In actual deployment, because of the influence of the different calculation modes of different blocks, the actual inference speed of a detection model is often not completely correlated with the FLOPs. From the results for Model (3) in Table VII, it can be seen that, after the introduction of the inference time index, although the parameter quantity and FLOPs are slightly increased compared with those of the previously obtained models, the inference time is further decreased, and the obtained model still maintains a good detection accuracy.

In the ablation experiments shown in Table VII, in the comparison of the results of Model (2) and Model (3), there is a detail worth discussing: Model (3) has a faster inference speed than Model (2), but its parameter quantity and FLOPs are slightly higher than those of Model (2). This is contrary to our intuition that, in general, models with lower FLOPs should be faster. To further investigate this phenomenon, we analyze the visualization of the SARNas search results in Fig. 8. More inverted residual blocks are selected in the search results of Model (3) than in those of Model (2). From the structure diagrams of the computational blocks in Fig. 3, it can be seen that, in contrast to the other two types of blocks (Ghost module and RepVGG), the inverted residual block uses depthwise separable convolution, which splits a common convolution into two depthwise convolutions. This causes the number of layers of the convolution calculation to increase, while the number of parameters and the amount of calculation are reduced. At the same time, because the current computing platforms used for AI inference (such as GPUs) exploit the parallel processing of large-scale data, the calculation capacity increases because each layer of calculation can be processed in parallel [65]. Therefore, the number of convolutional layers in the network becomes the dominant factor in determining the GPU inference time when there is little difference in model computation. This is the origin of the model performance difference between Model (3) and Model (2) in the ablation experiments. The experimental results reflect the necessity of considering the actual inference time of the model in NAS tasks, and they also prove the practicability of SARNas for real model deployment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

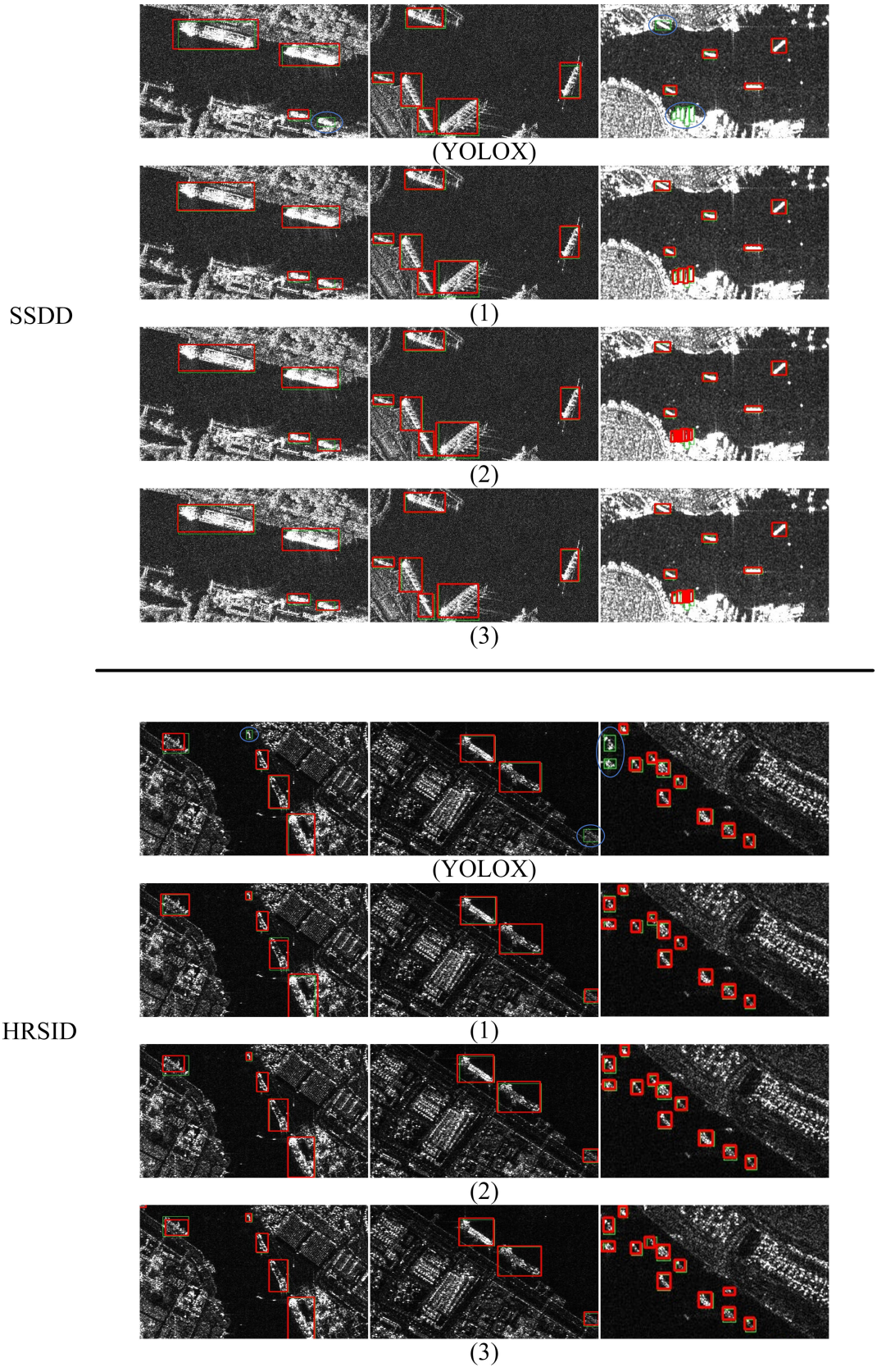


Fig. 7. Detection results from the ablation experiments. The images are labeled corresponding to the different ablated models for which the experimental results are given in Table VII. The green bounding boxes represent manual labeling information, and the red bounding boxes represent the model detection results. The blue circles represent missed targets in the detection results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

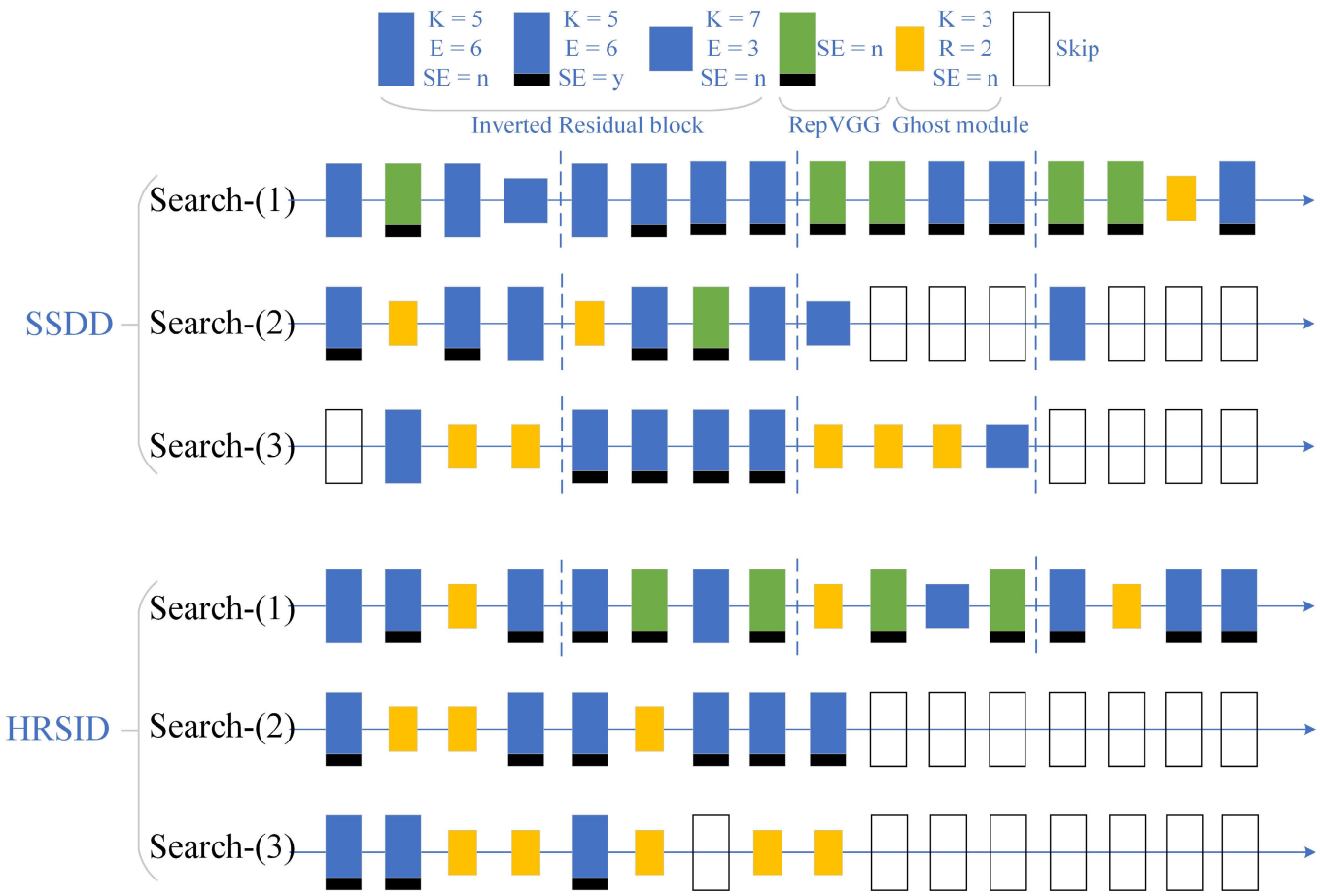


Fig. 8. Visual legend for the results of the ablation experiment, where Search-(1) means the result of a search only with reference to the accuracy factor, Search-(2) represents the search result under a constraint on the maximum amount of computation of the searched model, and Search-(3) means that, along with the accuracy and computation limit, the inference time is also considered as a factor during the search.

In summary, the above analysis of the results of the ablation experiments shows that our SARNas method can identify a network structure that is more suitable for the current SAR target detection task while simultaneously ensuring the accuracy and lightweight of the model. By considering the amount of model computation and the inference time to guide the search from the perspective of actual deployment, SARNas can obtain a network model that meets the hardware requirements of actual deployment (low computing power and fast inference speed) while ensuring its detection accuracy. The visual detection results of the models obtained in the ablation experiments are shown in Fig. 7.

2) Experiments on the Anchor-Based Detection Method YOLOV5: In the above experiments, we verified the effectiveness of our proposed SARNas method on the anchor-free algorithm YOLOX. To prove that our method can still achieve good results on other detection models, especially anchor-based models, we choose YOLOV5 [46], which is the most widely used anchor-based model, for additional experiments based on the same search space and search pipeline. The experimental results are shown in Tables VIII and IX.

As shown in Tables VIII and IX, compared with the middle-size YOLOV5 model (YOLOV5m), our obtained model requires only approximately one-twentieth of the FLOPs, and the inference speed is doubled, with only a slightly lower accuracy (accuracy drops of 0.2% on the SSDD dataset and 0.7% on the HRSID dataset). Compared with the tiny-size YOLOV5n model, our obtained model shows improvements in both accuracy (0.4% improvement on the SSDD dataset and 1.9% on the HRSID dataset) and inference speed, with less calculation. These results prove that our SARNas method also has excellent lightweight model structure search ability for anchor-based detection algorithms, as represented by YOLOV5.

3) Comparison With the Latest SAR Target Detection Methods: To prove that the models obtained by our search method are more suitable for SAR target detection tasks, we experimentally compare their performance with that of current mainstream target detection algorithms. To ensure fair comparisons, we divide these object detection algorithms into two categories according to whether they use anchor-free or anchor-based methods and compare their effects with those of the models obtained by our search method based on YOLOX and YOLOV5. The specific result comparisons are shown in Table VIII, and corresponding visual images of the detection results are shown in Fig. 9. From these comparisons, it can be seen that the models found by our method exceed other state-of-the-art detection algorithms in SAR target detection ability. In addition, the performance of the YOLOX-based SAR target detector automatically found by our SARNas method has been verified on large scenes, as shown in Fig. 10. The results show that our automatically found that the SAR target detector

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
COMPARISON WITH THE LATEST DETECTION METHODS AND SAR SHIP DETECTION METHODS ON SSDD

	Method	Params (M)	FLOPs (G)	mAP (%)	Average Inference Time (ms)
Anchor-free	CornerNet [47]	201.04	707.75	94.7	95.61
	CenterNet [48]	14.21	20.40	95.1	31.54
	FCOS [49]	60.97	78.63	95.3	48.67
	AFSar [34]	—	8.66	97.7	—
	LPEDet [38]	5.68	18.38	97.4	7.01
	CenterNet++ [52]	—	—	95.1	33.0
	FBR-Net [2]	32.5	—	94.1	40.10
	YOLOX [35]	12.45	41.78	96.2	11.30
	Ours (based on YOLOX)	**1.39**	5.45	97.8	5.54
Anchor-based	YOLOV3 [21]	61.52	77.54	96.2	45.81
	FPN [50]	63.56	71.65	96.5	78.30
	Faster R-CNN [18]	41.12	91.41	96.4	45.36
	Cascade R-CNN [51]	69.17	119.05	96.8	65.56
	YOLOV5m [46]	20.87	48.2	98.8	9.10
	YOLOV5n [46]	1.76	4.1	98.1	5.80
	Ours (based on YOLOV5)	2.31	**2.8**	98.5	**4.40**

has good applicability for target detection in large-scale SAR images.

To further prove that the structures found by our SARNas method have a better detection effect for SAR target detection, we collected and contrasted the latest SAR target detection algorithms based on CNN model designs. The relevant results are compared in Tables VIII and IX. These results show that, compared with other SAR target detection algorithms, the structures found by the SARNas method have better detection accuracy and faster detection speed on a challenging SAR target detection dataset.

4) Comparative Experiments on the Multiclass SAR Target Detection Dataset MSAR: To verify the effectiveness of our method on other kinds of SAR target detection tasks, we additionally conducted comparative experiments on the MSAR dataset. Because YOLOX and YOLOV5 are mainly used as the benchmark models for our SARNas method in these experiments, we mainly consider the YOLO series of models for comparison in this part of the experimental analysis. The experimental results are shown in Table X.

Compared with the original YOLOX model, the new YOLOX-based structure obtained through the SARNas search shows an increase of 5.7% in the detection accuracy index mAP, while the number of parameters is decreased by 76.6% and the computational complexity (in FLOPs) is decreased by 75.4%. Compared with the original control YOLOV5 model, for the new YOLOV5-based structure obtained by the SARNas method, the detection accuracy (mAP) is increased by 0.1%, the number of parameters is decreased by 68.5%, and the calculation amount (in FLOPs) is decreased by 67.7%.

Compared with smaller datasets, large-scale datasets contain more image samples and information content, so they are more sensitive to the model size (the number of parameters and the amount of computation). Therefore, on large-scale SAR target detection datasets, larger models have innate detection advantages over lightweight models. However, after structure optimization with our proposed SARNas method, even lightweight models can still achieve an excellent detection effect on large-scale SAR datasets. The detection effect on the MSAR dataset is visualized in Fig. 11.

5) Comparison With Traditional SAR Target Detection Methods: To verify the effectiveness of our SARNas method compared to traditional detection algorithms, we compare the performance of the models found through the SARNas search process with those of the CFAR-based detection algorithms CA-CFAR and OS-CFAR. The experimental results are shown in Table XI. Because the performance indicators of traditional detection algorithms are not the same as those of deep learning algorithms, we select three common evaluation indicators, P, R, and F1, to compare the detection performance.

From the experimental results in Table XI, it can be seen that the model structures found through the SARNas search achieve higher precision and F1 scores than the traditional detection algorithms. At the same time, algorithms based on deep learning can automatically learn appropriate model parameters for SAR target detection tasks. Therefore, compared with traditional algorithms, which require the manual design of the feature extraction process and threshold setting, our SARNas search results offer stronger adaptability and a higher degree of automation.

6) Verification of the Ability of Models Found Using SARNas to Extract Strong Scattering Points From SAR Targets: The imaging mechanisms of optical images and SAR images are fundamentally different. Specifically, SAR images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

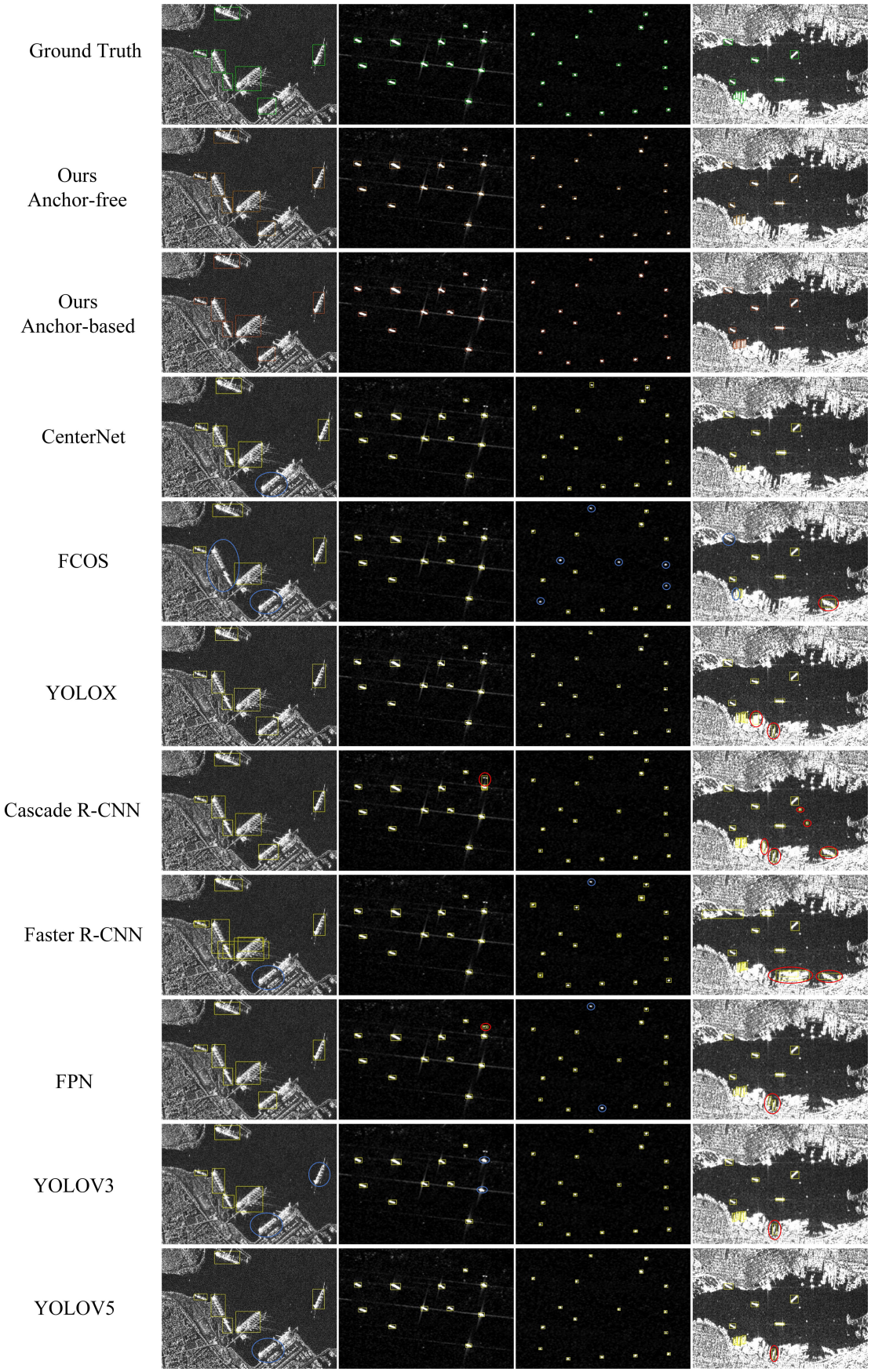


Fig. 9. Comparison with the latest detection methods. The green ground-truth bounding boxes are manual labeling information. Blue circles represent targets missed by the model, and red circles represent targets misdetected by the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

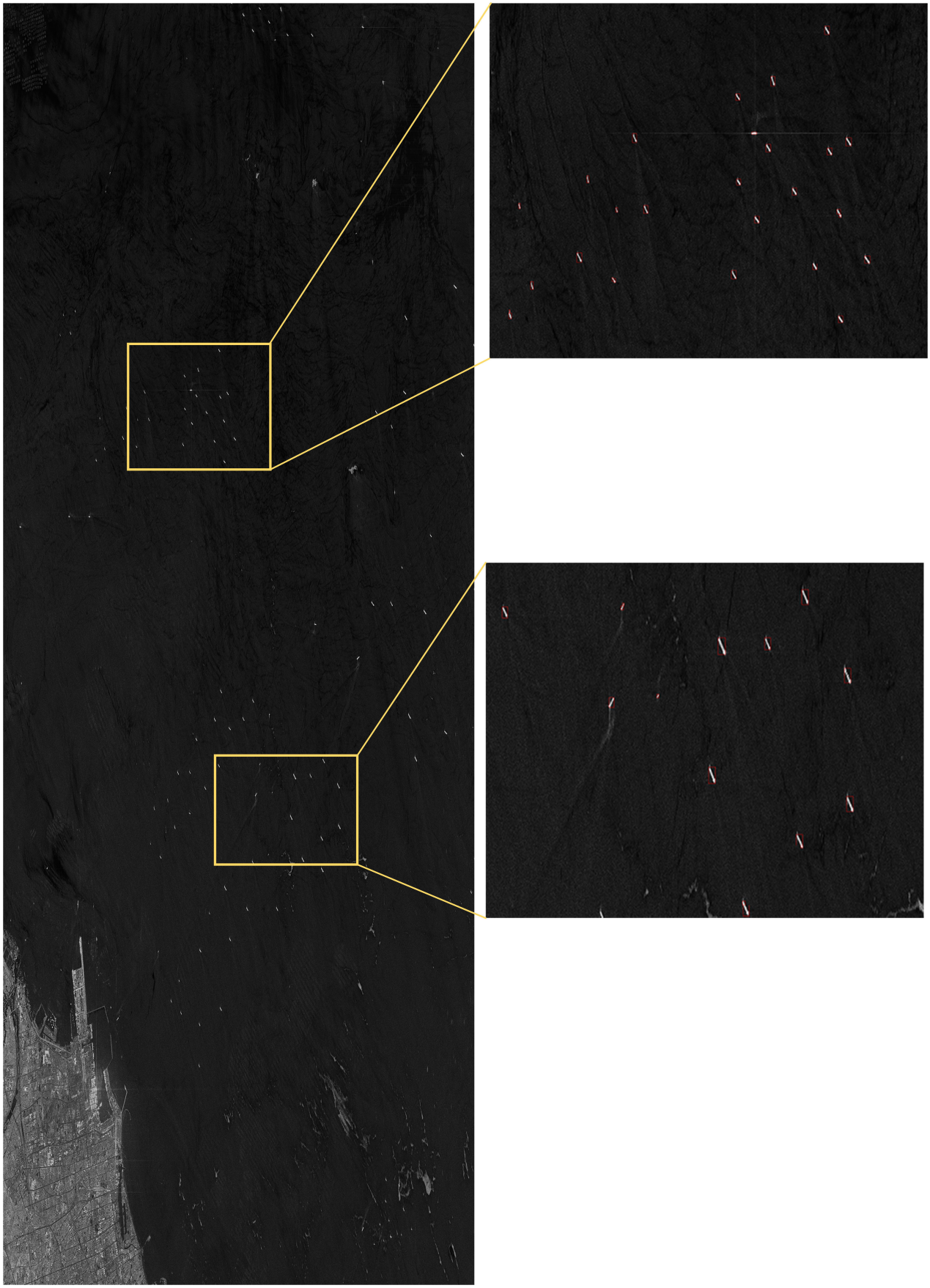


Fig. 10. Visualization of our algorithm on a large-scale image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
COMPARISON WITH THE LATEST DETECTION METHODS AND SAR SHIP DETECTION METHODS ON HRSID

	Method	Params (M)	FLOPs (G)	mAP (%)	Average Inference Time (ms)
Anchor-free	CenterNet2 [61]	71.6	—	64.5	—
	LPEDet [38]	5.68	18.38	89.7	7.01
	YOLOX [35]	12.45	41.78	87.3	19.30
	Ours (based on YOLOX)	**1.28**	5.46	89.1	**4.81**
Anchor-based	Faster R-CNN [18]	41.53	91.41	74.0	53.04
	SAR-net [53]	42.6	104.2	84.7	118.0
	YOLOV5m [46]	20.85	47.9	93.5	9.10
	YOLOV5n [46]	1.76	4.1	90.9	5.80
	Ours (based on YOLOV5)	1.99	**2.4**	92.8	4.9

TABLE X
COMPARISON WITH THE LATEST DETECTION METHODS ON MSAR

Method	Params (M)	FLOPs (G)	AP (%)				mAP (%)	Average Inference Time (ms)
			ship	oil tank	airplane	bridge		
YOLOV3 [21]	61.54	77.56	91.2	79.9	82.3	32.2	71.4	31.18
YOLOV4-tiny [22]	5.92	16.1	93.5	69.7	48.7	64.5	69.1	9.8
YOLOXs [35]	5.08	17.71	94.3	74.9	68.2	67.3	76.2	9.35
YOLOV5s [46]	7.02	15.8	96.9	93.4	86.6	89.7	91.7	6.5
Ours (based on YOLOX)	**1.19**	**4.36**	95.6	80.4	71.8	79.6	81.9	**4.7**
Ours (based on YOLOV5)	2.21	5.10	97.4	92.5	89.2	88.1	**91.8**	5.6

TABLE XI
COMPARISON WITH THE LATEST TRADITIONAL CFAR METHODS ON THE SSDD DATASET

Method	P (%)	R (%)	F1 (%)
CA-CFAR [58]	85.9	98.1	91.6
OS-CFAR [59]	84.2	98.5	90.2
Ours (based on YOLOX)	96.3	95.6	96.0
Ours (based on YOLOV5)	97.4	94.8	96.1

are obtained based on the processing of microwave signals transmitted and received by SAR. Different from optical remote sensing images, the information in an SAR image depends on the ability of the detection targets to reflect a radar beam, and the main information in the image is formed via backscattering by the target ground objects. This backscattering affects the gray values in the SAR image. In addition to the operating parameters of the radar system itself, the factors that affect backscattering are mainly related to the physical characteristics of the target, such as the roughness and complex permittivity of its surface. Therefore, differences in physical characteristics between the detected targets (ships, aircraft, and other objects) and the background (land and ocean) in SAR images lead to differences in their electromagnetic scattering characteristics for radar beams, which manifest as gray value differences in SAR images [63]. Moreover, the overall contours of targets in SAR images are typically significantly different from the background.

Different from the NAS methods used in optical remote sensing image processing, our method can select a neural model structure with better adaptability to the different electromagnetic scattering characteristics of SAR targets. Specifically, the combinations of network structures screened by the SARNas method can better extract the geometric contour features arising due to the special electromagnetic scattering characteristics of SAR targets in the feature map extraction stage.

To more intuitively show the ability of the model structures searched by the SARNas method to extract target strong scatterers in SAR images, we visualize examples of strong scatterers in SAR images by presenting the visual feature maps obtained in the actual reasoning process of the model. In Fig. 12, we use the Harris–Laplace corner detector [62] to extract and visualize strong scatterers in several SAR images, and it can be seen that these strong scatterers are mainly distributed around and within the corresponding SAR targets. Fig. 13 shows examples of the feature maps obtained during the inference process of the new detection network architecture obtained through a SARNas search when applied to SAR images of various categories. Specifically, in Fig. 13, we visualize the first eight channels of the feature maps extracted from the input SAR images in the backbone part of the detection network. As seen by comparing the results with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

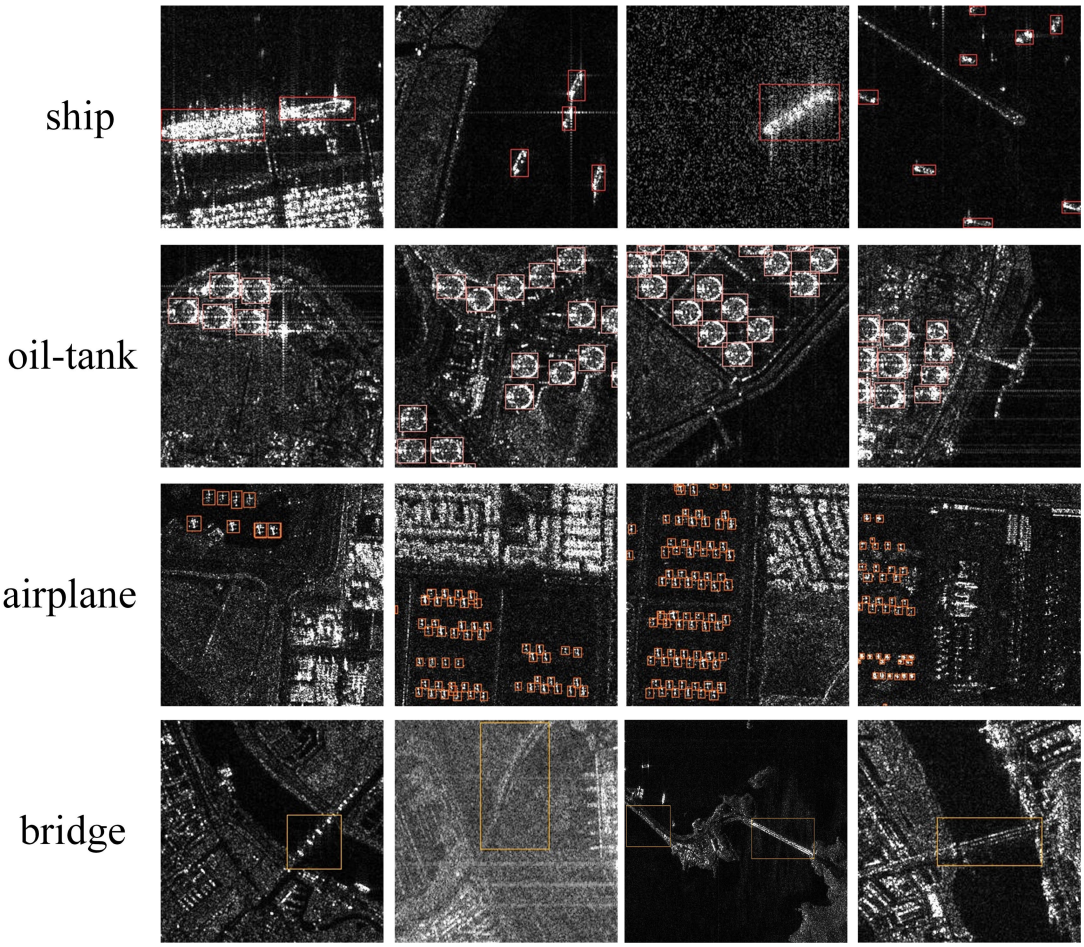


Fig. 11. Visualization of the detection results of our algorithm on the MSAR dataset.

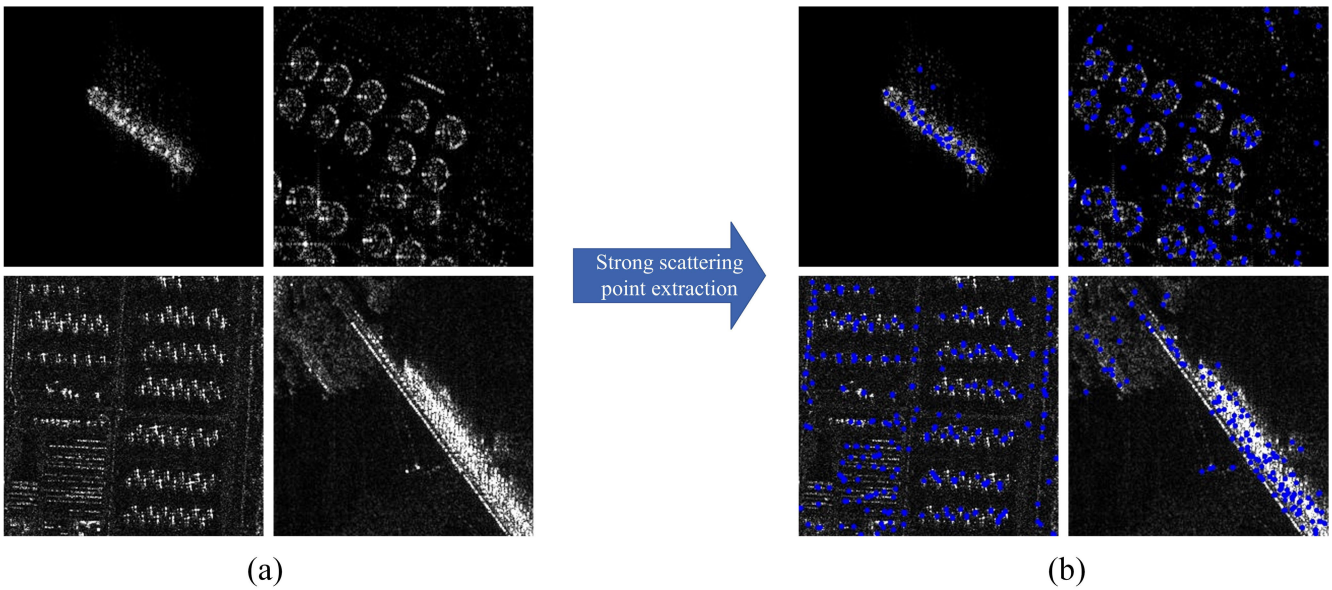


Fig. 12. Illustration of strong scattering point extraction for SAR targets using the Harris–Laplace corner detector. (a) Original SAR images. (b) Images after the strong scattering points in the SAR images are extracted and marked, where blue dots represent strong scattering points.

the strong scattering points extracted in Fig. 12, the detection network obtained by SARNas can accurately perceive the geometric contour information of the strong scattering points associated with SAR targets. On the basis of these geometric features, target recognition in SAR images can be quickly carried out. Thus, it can be proven that, compared with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

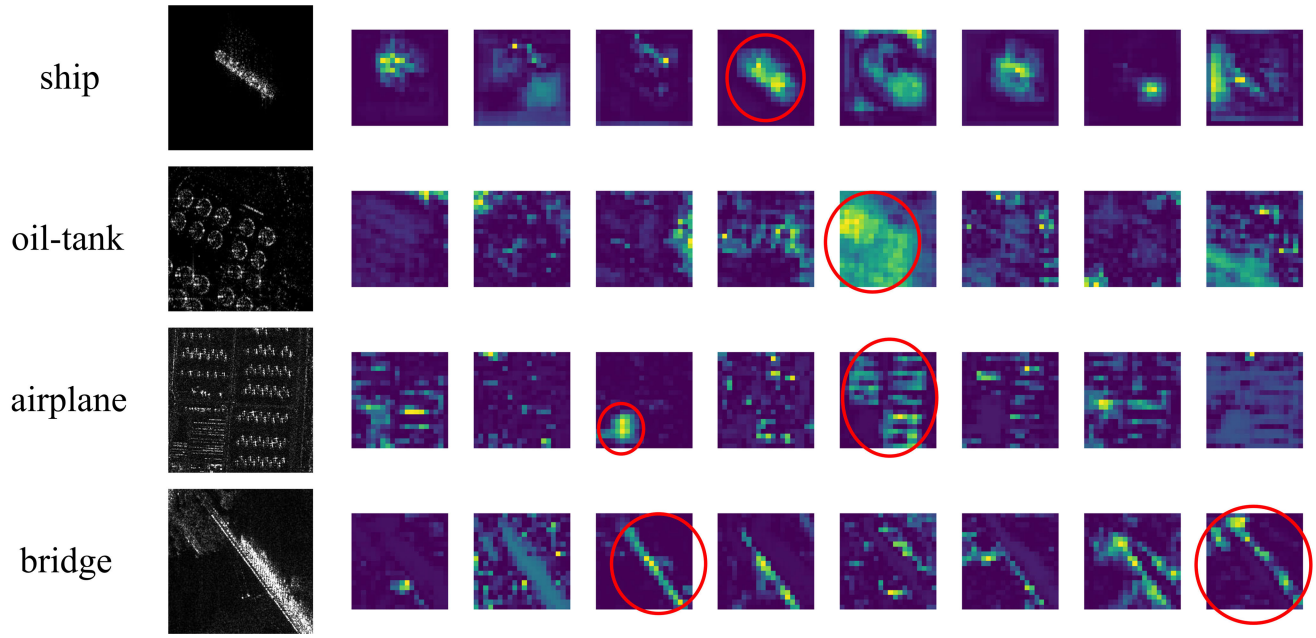


Fig. 13. Partial channel feature maps of the model obtained by SARNas in the inference process for SAR images. The red circles indicate the distribution areas of strong scatterers in the SAR images as detected by the model.

existing NAS methods for optical remote sensing images, our SARNas method has stronger perception and learning capabilities for SAR target characteristics and, thus, has better adaptability to SAR target detection tasks.

V. Conclusion

In this article, we propose SARNas, an NAS method for SAR target detection, which can search for the optimal backbone network structure for a selected benchmark target detection network. SARNas accounts for the detection accuracy, size, and inference speed of the model when searching for a network structure and, finally, finds an optimal structure suitable for the current SAR target detection task and the performance limitations of the deployment hardware.

We demonstrate the effectiveness of our approach through a series of experiments. We select two SAR target detection datasets, SSDD and HRSID, to conduct experiments on the YOLOX and YOLOV5 models. The experimental results show that, based on these existing models, our proposed SARNas method can search for SAR target detector structures with lighter weight, lower computational complexity, and a faster inference time while ensuring accuracy. The resulting SAR target detection models are more suitable for actual deployment on edge devices. We also report a set of ablation experiments conducted to verify the effectiveness of our proposed multiobjective optimization loss function-SARMI-Loss. The results of these ablation experiments show that our proposed SARMI-Loss plays an effective role in guiding the model search in terms of the inference speed, the number of parameters, and computational burden.

In addition, because of the ability of the NAS approach to design neural network architectures in accordance with specific tasks, this approach can also be beneficially applied to other SAR image processing tasks beyond target detection. For example, in previous work on optical-SAR registration using deep learning [60], the authors designed a network composed of fully connected convolutional layers for SAR image feature extraction and denoising, which achieved excellent registration results. Furthermore, in the future, researchers pursuing related work can also consider introducing the NAS approach to further optimize SAR image feature extraction networks and residual denoising networks for the optical-SAR registration task to reduce the workload of manual design and achieve a better registration effect. Similarly, the NAS approach can also be applied to optimize the model design workflow for few-shot SAR object detection. For example, in the work of Zhou et al. [64], the authors introduced a metalearning method to construct a lightweight few-shot SAR target detection algorithm. In this algorithm, DarknetS, a feature extraction network composed of multiple convolutional layer modules, is designed to enhance the feature representation of SAR images. Therefore, it would also be possible to consider the use of the NAS approach to assist in the design of this feature extraction network.

In summary, experimental verification results for our proposed SARNas method show that NAS technology can play a positive role in algorithm design for SAR target detection by effectively reducing the necessary manual effort. In the future, we will introduce an analysis of the SAR electromagnetic scattering mechanism into the NAS design process to guide the learning of physically interpretable SAR target detectors.

References

[1] L. Du, H. Dai, Y. Wang, W. Xie, and Z. Wang, "Target discrimination based on weakly supervised learning for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 461–472, Jan. 2020.

[2] J. Fu, X. Sun, Z. Wang, and K. Fu, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

[3] D. Gu and X. Xu, "Multi-feature extraction of ships from SAR images," in *Proc. 6th Int. Congr. Image Signal Process. (CISP)*, Hangzhou, China, Dec. 2013, pp. 454–458, doi: 10.1109/CISP.2013.6744037.

[4] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.

[5] B. Hou, X. Chen, and L. Jiao, "Multilayer CFAR detection of ship targets in very high resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 811–815, Apr. 2015.

[6] W. Yu, Y. Wang, H. Liu, and J. He, "Superpixel-based CFAR target detection for high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 730–734, May 2016, doi: 10.1109/LGRS.2016.2540809.

[7] T. Li, Z. Liu, R. Xie, and L. Ran, "An improved superpixel-level CFAR detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 184–194, Jan. 2018, doi: 10.1109/JSTARS.2017.2764506.

[8] Y. Huang and F. Liu, "Detecting cars in VHR SAR images via semantic CFAR algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 801–805, Jun. 2016, doi: 10.1109/LGRS.2016.2546309.

[9] D. Charalampidis and G. W. Stein, "Target detection based on multiresolution fractal analysis," *Proc. SPIE*, vol. 6567, May 2007, Art. no. 65671B, doi: 10.1117/12.720145.

[10] E. Abdelkawy and D. McGaughy, "Wavelet-based image target detection methods," *Autom. Target Recognit.*, vol. 5094, pp. 1–11, Sep. 2003, doi: 10.1117/12.487139.

[11] S. Chen and X. Li, "A new CFAR algorithm based on variable window for ship target detection in SAR images," *Signal, Image Video Process.*, vol. 13, no. 4, pp. 779–786, Jun. 2019.

[12] S. Wang, M. Wang, S. Yang, and L. Jiao, "New hierarchical saliency filtering for fast ship detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 351–362, Jan. 2017.

[13] P. Iervolino and R. Guida, "A novel ship detector based on the generalized-likelihood ratio test for SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3616–3630, Aug. 2017.

[14] H. Lang, Y. Xi, and X. Zhang, "Ship detection in high-resolution SAR images by clustering spatially enhanced pixel descriptor," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5407–5423, Aug. 2019.

[15] X. Leng, K. Ji, X. Xing, S. Zhou, and H. Zou, "Area ratio invariant feature group for ship detection in SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2376–2388, Jul. 2018.

[16] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[17] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1440–1448, doi: 10.1109/ICCV.2015.169.

[18] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[19] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 779–788, doi: 10.1109/CVPR.2016.91.

[20] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6517–6525, doi: 10.1109/CVPR.2017.690.

[21] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[22] C.-Y. Wang, A. Bochkovskiy, and H. M. Liao, "Scaled-YOLOv4: Scaling cross stage partial network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13024–13033.

[23] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.*, vol. 9905. Cham, Switzerland: Springer, 2016, pp. 21–37.

[24] R. Wang et al., "An improved faster R-CNN based on MSER decision criterion for SAR image ship detection in harbor," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 1322–1325, doi: 10.1109/IGARSS.2019.8898078.

[25] Y. Q. Liang, X. H. Wang, and L. F. Chen, "SAR target detection method based on deep learning," *Radar Sci. Technol.*, vol. 17, no. 5, pp. 579–586, 2019.

[26] Z. Wang, L. Du, J. Mao, B. Liu, and D. Yang, "SAR target detection based on SSD with data augmentation and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 150–154, Jan. 2019.

[27] H. Li, N. Wang, M. Zhu, X. Yang, and X. Gao, "Recent advances in neural architecture search: A survey," *J. Softw.*, vol. 33, no. 1, pp. 129–149, 2022. [Online]. Available: http://www.jos.org.cn/1000-9825/6306.htm

[28] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8697–8710.

[29] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," 2018, *arXiv:1806.09055*.

[30] X. Wang, J. Lin, J. Zhao, X. Yang, and J. Yan, "EAutoDet: Efficient architecture search for object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Tel Aviv, Israel, Oct. 2022, pp. 668–684.

[31] B. Wu et al., "FBNet: Hardware-aware efficient ConvNet design via differentiable neural architecture search," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 10726–10734, doi: 10.1109/CVPR.2019.01099.

[32] M. Tan et al., "MnasNet: Platform-aware neural architecture search for mobile," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 2815–2823, doi: 10.1109/CVPR.2019.00293.

[33] H. Cai, L. Zhu, and S. Han, "ProxylessNAS: Direct neural architecture search on target task and hardware," 2018, *arXiv:1812.00332*.

[34] H. Wan et al., "AFSar: An anchor-free SAR target detection algorithm based on multiscale enhancement representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219514, doi: 10.1109/TGRS.2021.3137817.

[35] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[36] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[37] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[38] Y. Feng et al., "A lightweight position-enhanced anchor-free algorithm for SAR ship detection," *Remote Sens.*, vol. 14, no. 8, p. 1908, Apr. 2022, doi: 10.3390/rs14081908.

[39] H. Pham, M. Y. Guan, B. Zoph, Q. V. Le, and J. Dean, "Efficient neural architecture search via parameter sharing," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2018, pp. 4095–4104.

[40] H. Zhu, S. Guo, W. Sheng, and L. Xiao, "SBNN: A searched binary neural network for SAR ship classification," *Appl. Sci.*, vol. 12, no. 14, p. 6866, Jul. 2022, doi: 10.3390/app12146866.

[41] Z. Li, L. Chen, X. Huang, Z. Zhang, and H. Shi, "Differentiable neural architecture search for SAR image ship object detection," in *Proc. IET Int. Radar Conf. (IET-IRC)*, vol. 2020, Nov. 2020, pp. 950–954, doi: 10.1049/icp.2021.0794.

[42] P. Zhang, J. Tang, H. Zhong, M. Ning, D. Liu, and K. Wu, "Self-trained target detection of radar and sonar images using automatic deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4701914, doi: 10.1109/TGRS.2021.3096011.

[43] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 4510–4520, doi: 10.1109/CVPR.2018.00474.

[44] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13728–13737, doi: 10.1109/CVPR46437.2021.01352.

[45] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 1577–1586, doi: 10.1109/CVPR42600.2020.00165.

[46] Ultralytics. *YOLOv5*. Accessed: 2021. [Online]. Available: https://github.com/ultralytics/yolov5

[47] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, Sep. 2018, pp. 734–750.

[48] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[49] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.

[50] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 936–944.

[51] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 6154–6162.

[52] H. Guo, X. Yang, N. Wang, and X. Gao, "A CenterNet++ model for ship detection in SAR images," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107787.

[53] S. Gao, J. M. Liu, Y. H. Miao, and Z. J. He, "A high-effective implementation of ship detector for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4019005, doi: 10.1109/LGRS.2021.3115121.

[54] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020, doi: 10.1109/ACCESS.2020.3005861.

[55] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021, doi: 10.3390/rs13183690.

[56] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-Excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020, doi: 10.1109/TPAMI.2019.2913372.

[57] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with Gumbel-Softmax," 2016, *arXiv:1611.01144*.

[58] C. Kuang, C. Wang, B. Wen, Y. Hou, and Y. Lai, "An improved CA-CFAR method for ship target detection in strong clutter using UHF radar," *IEEE Signal Process. Lett.*, vol. 27, pp. 1445–1449, 2020.

[59] E. Hyun and J.-H. Lee, "A new OS-CFAR detector design," in *Proc. 1st ACIS/JNU Int. Conf. Comput., Netw., Syst. Ind. Eng.*, May 2011, pp. 133–136.

[60] D. Xiang, Y. Xie, J. Cheng, Y. Xu, H. Zhang, and Y. Zheng, "Optical and SAR image registration based on feature decoupling network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5235913, doi: 10.1109/TGRS.2022.3211858.

[61] X. Zhou, V. Koltun, and P. Krähenbühl, "Probabilistic two-stage detection," 2021, *arXiv:2103.07461*.

[62] E. Rosten and T. Drummond, "Machine learning for high-speed corner detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2006, pp. 430–443.

[63] Z. Zhou et al., "HRLE-SARDet: A lightweight SAR target detection algorithm based on hybrid representation learning enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5203922, doi: 10.1109/TGRS.2023.3251694.

[64] Z. Zhou et al., "FSODS: A lightweight metalearning method for few-shot object detection on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5232217, doi: 10.1109/TGRS.2022.3192996.

[65] P. K. A. Vasu, J. Gabriel, J. Zhu, O. Tuzel, and A. Ranjan, "MobileOne: An improved one millisecond mobile backbone," 2022, *arXiv:2206.04040*.

[66] J. Chen et al., "Large-scale multi-class SAR image target detection dataset-1.0[OL]," *J. Radars*, no. 1, 2022. [Online]. Available: https://radars.ac.cn/web/data/getData?dataType=MSAR



Wentian Du received the B.S. degree in electronic information engineering from Anhui University, Hefei, Anhui, China, in 2021, where he is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, School of Electronics and Information Engineering.

His research interests mainly focus on synthetic aperture radar (SAR) image processing and deep learning in remote sensing applications.



Jie Chen (Member, IEEE) received the Ph.D. degree in computer science and technology from the Hefei Institute of Physical Science, Chinese Academy of Sciences, University of Science and Technology of China, Hefei, China, in 2019.

He is currently an Associate Professor with Anhui University, Hefei, where he is also a Master Supervisor of the Information Materials and Intelligent Sensing Laboratory of Anhui Province and the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education. He is also a Post-Doctoral Fellow with the 38th Research Institute, China Electronics Technology Group Corporation, Hefei. He has presided over ten projects, including a National Natural Science Foundation Youth Project, an Anhui Provincial Natural Science Foundation Youth Project, and a China Postdoctoral Science Foundation General Project. He has published dozens of papers in top journals/conference proceedings, such as IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and *Research*. Patents for more than ten of his inventions have been accepted and authorized. His research interests include deep learning, intelligent interpretation of remote sensing images, and autonomous driving.

Dr. Chen won the Silver Medal of Anhui Province in the First National Postdoctoral Innovation and Entrepreneurship Competition in 2021 and was awarded the honorary title of Excellent Postdoctoral Fellow.



Chaochen Zhang received the B.E. degree in electronic information engineering from Anhui University, Hefei, Anhui, China, in 2022, where he is currently pursuing the M.E. degree with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, School of Electronics and Information Engineering.

His research interests mainly focus on the intelligent interpretation of synthetic aperture radar (SAR) images.



Po Zhao received the B.S. degree in communication engineering from Zhengzhou University, Zhengzhou, Henan, China, in 2022. He is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, Anhui University, Hefei, China.

His research interests mainly focus on synthetic aperture radar (SAR) image processing and deep learning in remote sensing applications.



Huiyao Wan received the M.S. degree from Anhui University, Anhui, China, in 2022, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education.

His research interests mainly focus on synthetic aperture radar (SAR) image processing and deep learning in remote sensing applications.



Zheng Zhou was born in Chuzhou, China, in 1996. He received the B.S. degree from the Information Materials and Intelligent Sensing Laboratory of Anhui Province and the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, School of Electronics and Information Engineering, Anhui University, Hefei, Anhui, China, in 2020, where he is currently pursuing the M.S. degree.

He is also an Intern at the Earth Observation R&D Center, 38th Research Institute, China Electronics Technology Group Corporation, Hefei. He has published several papers in high-level journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, a top journal in the field of remote sensing. His research interests include few-shot object detection in synthetic aperture radar (SAR) images and object detection in multipolarization SAR images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.



Yice Cao received the Ph.D. degree from the Remote Sensing Image Processing and Fusion Group, School of Electronic Engineering, Xidian University, Xi'an, China, in 2022.

She is currently a Lecturer with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, Anhui University, Hefei, China. Her primary research focuses on the intelligent interpretation of remote sensing images and deep learning.



Zhixiang Huang (Senior Member, IEEE) was born in Anhui, China. He received the B.S. and Ph.D. degrees in computational electromagnetics from Anhui University (AHU), Hefei, China, in 2002 and 2007, respectively.

Since 2008, he has been a Full Professor with the School of Electronic Information and Engineering, AHU. From September 2010 to September 2011, he was a Visiting Scholar with Iowa State University, Ames, IA, USA. From August 2013 to October 2013, he was a Visiting Professor with The University of Hong Kong, Hong Kong. From February 2014 to February 2015, he was a Visiting Professor with the Beijing National Laboratory for Condensed Matter Physics, Institute of Physics, Chinese Academy of Sciences, Beijing, China. He has coauthored one monograph and three book chapters for CRC Press, Boca Raton, FL, USA, and InTech Publishers, London, U.K. He has coauthored 80 peer-reviewed journal articles included in the Web of Science Core Collection. His research interests include theoretical and computational research in electromagnetics and imaging, focusing on multiphysics and interdisciplinary research, and fundamental and applied aspects of metamaterials and active metamaterials.

Dr. Huang is also a member of The Optical Society of America (OSA). He was awarded by the National Science Foundation for Outstanding Young Scholars of China in 2017. He was a recipient of the Changjiang Distinguished Professorship of the Ministry of Education in 2020. He was a recipient of the Second Prize of Science and Technology from the Anhui Provincial Government, China, in 2015.



Yingsong Li (Senior Member, IEEE) received the B.S. degree in electrical and information engineering and the M.S. degree in electromagnetic field and microwave technology from Harbin Engineering University (HEU), Harbin, China, in 2006 and 2011, respectively, and the Ph.D. degree from the Kochi University of Technology (KUT), Kochi, Kerela, and HEU in 2014.

He has been a Full Professor with the School of Electronic and Information Engineering, Anhui University, Hefei, China, since March 2022. He was a Professor with HEU from 2014 to 2022, a Visiting Scholar with the University of California at Davis, Davis, CA, USA, from March 2016 to March 2017, and a Visiting Professor with the University of York, York, U.K., in 2018. He has been a Visiting Professor with Far Eastern Federal University (FEFU), Vladivostok, Russia, and KUT. Since 2018, he has been a Visiting Professor with the School of Information, KUT. He was a Post-Doctoral Researcher with the Key Laboratory of Microwave Remote Sensing, Chinese Academy of Sciences, Beijing, China, from 2016 to 2020. He has authored or coauthored approximately 300 journal articles and conference papers in various areas of electrical engineering. His research interests include remote sensing, underwater communications, signal processing, adaptive filters, metasurface design, and microwave antennas.

Dr. Li is a fellow of the Applied Computational Electromagnetics Society (ACES Fellow) and a Senior Member of the Chinese Institute of Electronics (CIE). He was the TPC Co-Chair of the 2019 IEEE International Workshop on Electromagnetics (iWEM) from 2019 to 2020, the 2019 IEEE 2nd International Conference on Electronic Information and Communication Technology (ICEICT 2019), the 2019 International Applied Computational Electromagnetics Society (ACES) Symposium-China, and the 2019 Cross Strait Quad-regional Radio Science and Wireless Technology Conference (2019 CSQRWC), and the TPC Chair of ICEICT from 2021 to 2022. He was the General Co-Chair of ICEICT 2020 and the General Chair of the IEEE 9th International Conference on Computer Science and Network Technology (ICCSNT 2021) and ICCSNT 2022. He also serves as the Session Chair or an Organizer for many international and domestic conferences, including IEEE Wireless Communications and Networking Conference (WCNC), IEEE Antennas and Propagation Society (AP-S), and ACES. He acts as a reviewer for numerous IEEE, IET, and Elsevier journals and other international journals. He has served as an Area Editor for *International Journal of Electronics and Communications* (AEÜ) from 2017 to 2020. He is an Associate Editor of IEEE ACCESS, the *Applied Computational Electromagnetics Society Journal* (ACES), and the *Alexandria Engineering Journal*.



Bocai Wu was born in Xiaogan, Hubei, China.

He is currently a Researcher-Level Senior Engineer with the 38th Research Institute, China Electronics Technology Group Corporation, Hefei, China. As the person in charge, he has presided over more than ten key radar projects. He has been granted ten invention patents and has published more than 20 high-level papers. His main research directions include artificial intelligence technology and applications, reconnaissance and surveillance radar system design, and synthetic aperture radar (SAR) applications.

Dr. Wu won nine provincial and ministerial-level scientific and technological progress awards in China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:24:04 UTC from IEEE Xplore. Restrictions apply.